

ColonyGame: Complete Code Flow & Optimization Audit

Script-to-script runtime map, generated dependency catalog, visual flowcharts, and prioritized improvement plan

Repository snapshot: 18 June 2026 • 229 executable/configuration surfaces • 50,991 indexed lines

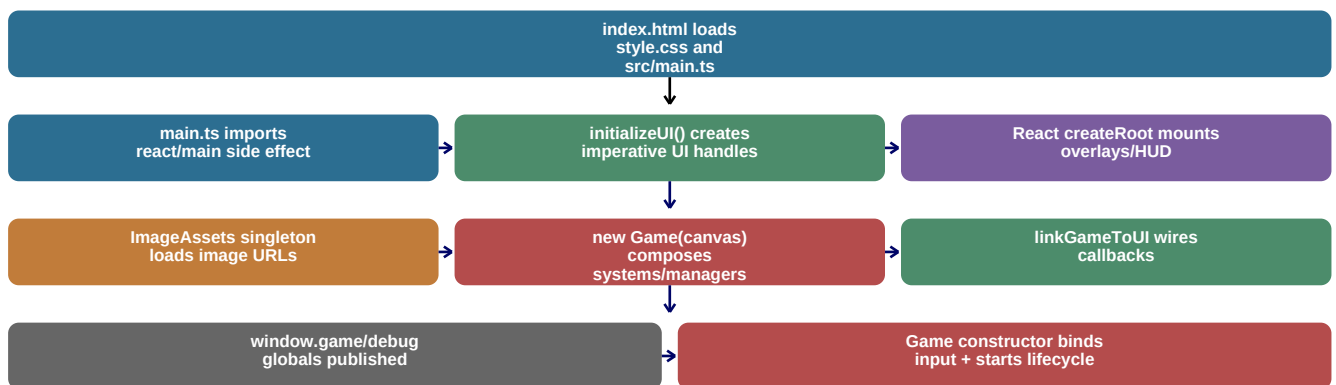
Scope statement: This report traces browser bootstrap, React and imperative UI, game-loop orchestration, simulation, AI/FSM, work assignment, navigation, combat, health, economy, events, rendering, audio, assets, build and deployment. Every TypeScript/TSX script is listed in the appendix with direct imports, exports, declarations, detected side effects, size and checksum. Binary assets are described as manifest-driven inputs rather than treated as scripts.

1. How to Read This Audit

- Solid left-to-right boxes indicate synchronous call/dependency progression; vertical arrows indicate transition to the next lifecycle phase.
- “Game” is both the composition root and a compatibility façade. Many managers receive the Game instance and therefore can access broad mutable state.
- The catalog is a static dependency map, not a dynamic profiler. Conditional branches, callbacks and state transitions are explained in subsystem notes; runtime frequency claims should be confirmed with performance traces.
- Type-only imports vanish at build time. React stores and bridges form an imperative interoperability layer between the canvas engine and React rendering.

2. End-to-End Bootstrap Flow

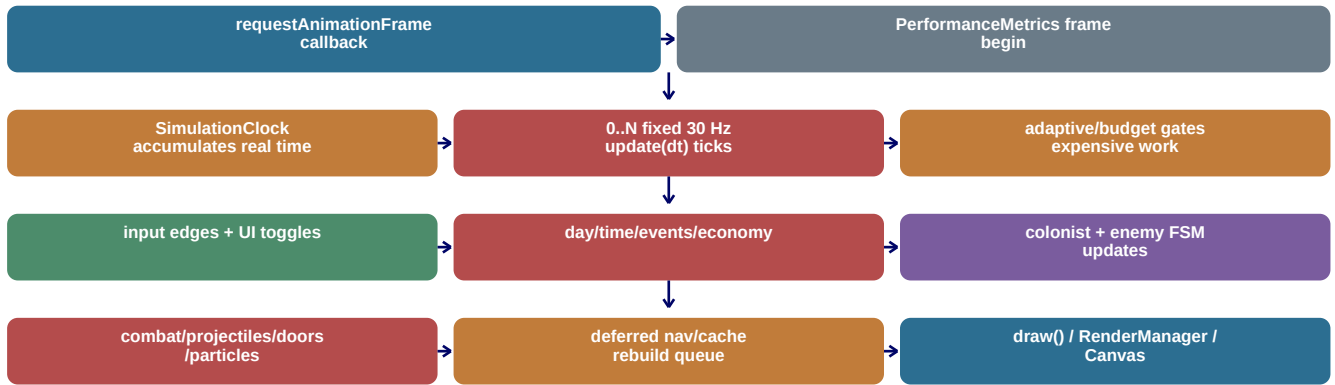
Browser startup and ownership handoff



- index.html supplies #root and #react-root and executes src/main.ts as an ES module. Vite resolves TypeScript, TSX, image and audio imports.
- main.ts initializes the legacy/imperative UI first, creates a loading overlay, awaits ImageAssets.loadAssets(), constructs Game only after image loading settles, then links UI handles and exposes debugging globals. Asset failure is non-fatal; canvas creation failure is fatal.
- Importing react/main.tsx mounts React immediately. React components subscribe to module-level stores. Imperative bridge functions update those stores from Game and legacy UI code.
- Game construction instantiates core state, time, camera, resources, navigation, rendering, combat, audio, performance, tutorial, event, inventory, work and medical collaborators; establishes terrain/entities; binds events; initializes debug/performance UI; and enters the frame lifecycle.

3. Main Frame / Simulation Flow

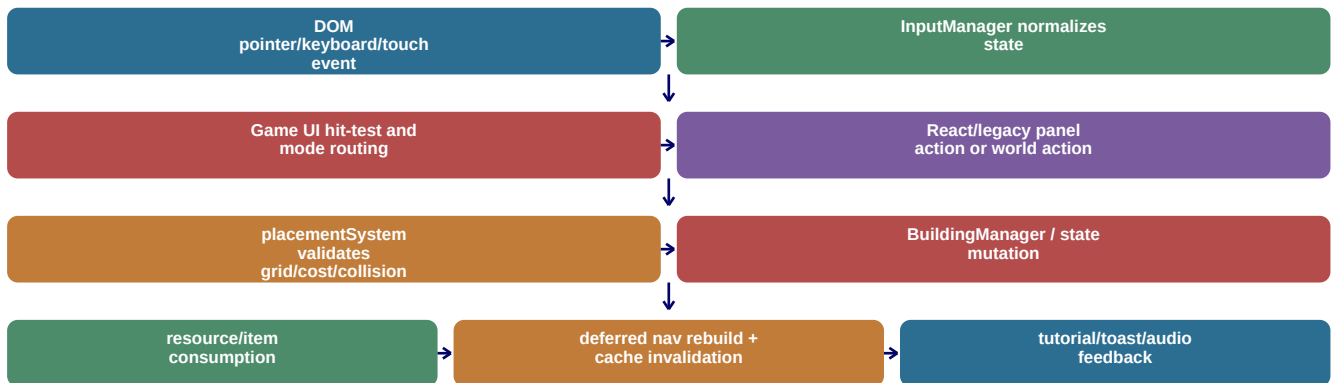
Frame lifecycle and fixed-step simulation



- **SimulationClock** decouples simulation time from display refresh. `Game.update` consumes a fixed `dt`, while `draw` is display paced. Pausing and speed multipliers are coordinated through `TimeSystem/Game` compatibility properties.
- **InputManager** records pointer, key and gesture state; `Game.bindInput` routes gestures and clicks to UI first, then selection, commands, placement, context menus and camera actions.
- **AdaptiveTickRate** and **BudgetedExecution** are intended to lower AI cadence and spread heavy work. Their effectiveness depends on every expensive subsystem honoring those gates; direct loops in `Game.update` can bypass them.
- Deferred rebuild processing orders render-cache invalidation before a full navigation rebuild, avoiding repeated immediate rebuilds during placement/mining batches.
- **RenderManager** orchestrates world rendering while specialized render modules draw terrain, buildings, items, colonists, weapons, particles, lighting, debug layers and HUD/UI. `DirtyRectTracker` and `RenderCache` reduce repeated work, but the current mixed canvas/DOM/React stack can still trigger redundant updates.

4. Player Input to World Mutation

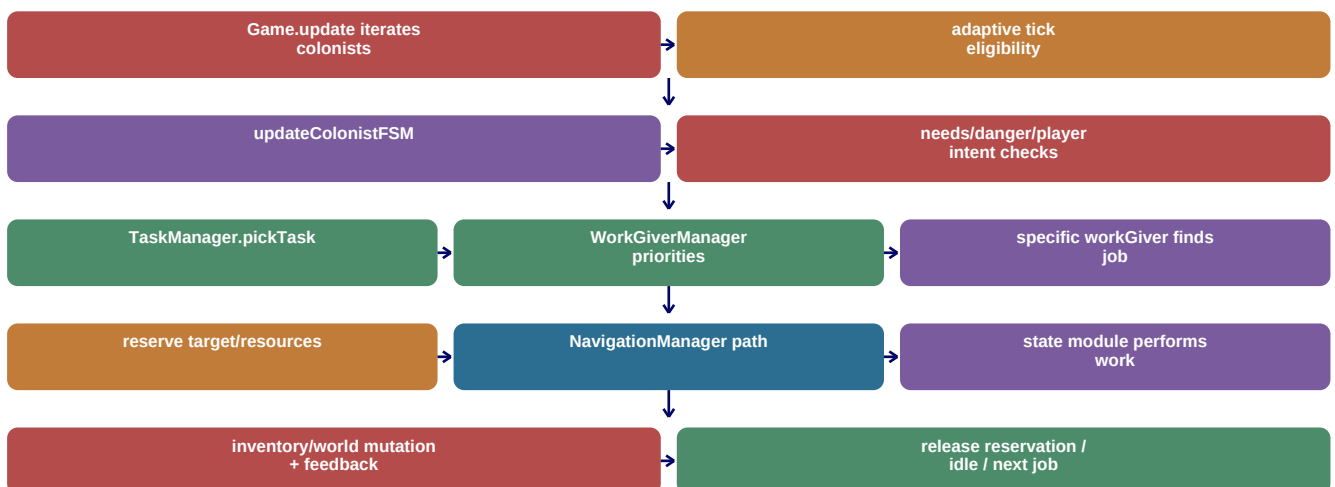
Command and construction path



- Coordinates pass through screen → camera/world conversion, then tile snapping where placement requires it. Touch placement uses a pending ghost and explicit nudge/rotate/confirm actions.
- Placement validates BUILD_TYPES metadata, map bounds, terrain, overlap, path/wall paint semantics and affordability. Successful construction creates an unfinished Building, consumes stockpiled resources, initializes special systems (doors/inventory), and creates construction work.
- Construction work gives discover unfinished buildings, ReservationManager prevents duplicate ownership, TaskManager sets the colonist task, navigation computes an approach path, and buildState advances work until BuildingManager completion effects execute.
- World mutation must invalidate pathfinding and relevant render caches. The deferred queue is the central anti-stutter mechanism; direct rebuildNavGridImmediate remains an escape hatch and should be rare.

5. Colonist AI and Work Assignment

Colonist task selection and state execution



- The FSM is the behavioral hub. It handles emergency/need states and delegates extracted state logic for building, cooking, cooling, equipment, guarding, mining, movement, research, sleep/rest, smelting, smithing and stonecutting. Remaining states still reside in the large switch.
- WorkPriority maps job types to RimWorld-like priorities. WorkGiverManager consults enabled work types and specific work-giver modules for construction, cooking, cooling, equipment, floor hauling, growing, hauling, medical, mining, patient behavior, plant cutting, research, smelting, smithing, stonecutting, tree planting and water collection.
- Reservations are correctness-critical. Every state exit—success, cancellation, incapacitation, drafted override, unreachable target or deletion—must release target/build/sleep/item reservations and clear paths. Game includes cleanup paths, but broad state mutation makes leaks possible.
- Player command intents must preempt ordinary work deterministically without being overwritten by the next AI scan. Drafted/guard/combat paths are separate high-priority control modes.

6. Navigation, Terrain and Doors

- Terrain generation creates the tile grid and mountain/ore data. navGrid translates terrain and buildings into walkability/cost cells. NavigationManager owns rebuild, blocked checks, approach points and path requests; core/pathfinding.ts implements A* grid search.
- ColonistNavigationManager and ColonistMovementManager consume paths, apply tile/floor speed multipliers, detect arrival and handle collision/door interaction. Enemy FSM uses the same world constraints but distinct target/attack behavior.
- DoorSystem adds dynamic blocking and queues. A mover detects a blocking door, requests opening, waits while necessary, passes when non-blocking, then releases queue participation. Door animation changes collision semantics, so stale nav/collision state can cause oscillation.
- Mining mutates terrain tiles and ore/resource outputs, marks assignments complete, and requests partial/full navigation and render updates. Zone systems represent stockpile, tree-growing and mining rectangles and feed both AI queries and rendering.

7. Economy, Items, Buildings and Production

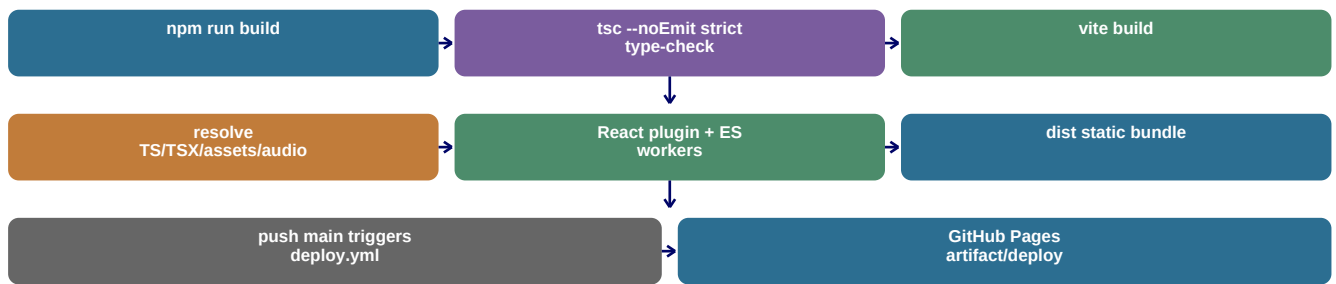
- ResourceSystem retains legacy aggregate counters/capacity. ItemManager and floorItems represent physical stacks and stockpile storage. Game synchronizes aggregate HUD totals from stockpiles and can materialize legacy resources into physical items. This dual model is a migration seam and a consistency risk.
- itemDatabase is the authoritative item catalog. Building inventory provides slot/capacity storage for specialized structures. ItemRenderer and floor-item rendering visualize world stacks. HaulManager/work givers move stacks into acceptable stockpile or building destinations.
- BUILD_TYPES defines cost, dimensions, category and behavior metadata. BuildingManager centralizes completion/removal and special effects, while placementSystem handles creation. BuildingUtilities and inventory helpers serve queries and transactions.
- Production chains are state/work-giver pairs: cooking (ingredients → stove work → bread storage), research (bench assignment → progress → unlock), stonecutting, smithing, smelting, growing/harvest and water collection. Each chain crosses work priority, reservation, navigation, inventory and feedback layers.

8. Combat, Health, Events, Audio and UI

- CombatManager coordinates high-level combat. pawnCombat chooses/executes colonist attacks; enemyFSM handles enemy behavior; combatSystem updates turrets and projectiles; weaponStats supplies data; unarmedCombat handles fallback melee. Particle modules create muzzle flashes, trails and impacts; weapon audio maps attacks to manifest keys.
- HealthSystem owns injuries, bleeding and overall health. MedicalWorkGiver identifies patients; MedicalManager and medical state paths reserve doctors/beds/items; medicalSystem defines treatments; surgerySystem resolves operations and hospital modifiers. Death/downing feeds Game over and task cleanup.
- EventManager schedules event definitions from events/index. Incidents include attacks, spoilage, cache windfalls, social/festival stories, solar flares and sandstorms. Events mutate state and communicate via messages/audio/render effects.
- AudioManager is a singleton over the audio manifest. Helper modules translate semantic actions into keys; MusicManager selects day/raid/game-over music. Browser autoplay rules require user-gesture resume and robust handling of unavailable assets.
- UI is hybrid: canvas/imperative modules coexist with React components. Stores expose subscribe/get/set-style state; bridges let Game update React. Duplicate legacy and modern menus/panels increase routing, z-order, input-capture and synchronization complexity.

9. Build and Deployment Flow

Developer and CI pipeline



- package.json defines dev, build and preview only; there is no test, lint, format, dependency-audit or PDF regeneration command.
- tsconfig uses strict mode, isolated modules, bundler resolution and noEmit. Vite targets esnext, includes audio extensions, and derives base from BASE_PATH for Pages.
- The deployment workflow is the only back end: this is a client-only static application. There is no game server, persistence API, database, authentication or authoritative multiplayer service. “Back-end performance” therefore means build/deploy efficiency today and architecture guidance for any future service.

10. Prioritized Improvement Plan

P0 Correctness & observability

Add deterministic unit tests for resource/item transactions, reservation release, FSM transitions, placement validity, door queues, health outcomes and pathfinding. Add seeded RNG and simulation replay fixtures. Instrument per-subsystem CPU time, queue depth, A* expansions, cache hit rate, entity counts and long tasks.

P0 Single source of truth

Finish migration from aggregate ResourceSystem counters to physical ItemManager/stockpile transactions, or define an explicit transactional ledger. Remove bidirectional “sync” ambiguity and assert conservation of items after every transaction in development builds.

P0 Decompose Game.ts

Turn Game into a narrow composition root. Extract input routing, lifecycle/frame scheduling, world generation, zone editing, command dispatch and compatibility properties. Replace manager access to the whole Game object with typed dependency interfaces and commands/events.

P1 AI performance

Use spatial indexes for nearest-job/target queries; cache work-giver candidate sets by dirty region; stagger need scans; cap path requests per tick; deduplicate identical route requests; and expose cancellation/timeout semantics. Profile before changing tick rates.

P1 Navigation

Adopt hierarchical/sector pathfinding or flow fields for common destinations, partial dirty-tile nav updates, path cache invalidation by nav revision, and a priority queue optimized for A*. Keep doors as dynamic edge costs rather than forcing broad rebuilds.

P1 Rendering

Use layered offscreen canvases for static terrain/floors, viewport culling for every entity/item/particle, sprite atlases, object pooling, DPR/zoom-aware cache keys and measured dirty-rectangle thresholds. Avoid full-canvas redraw when static layers are unchanged.

P1 UI workflow

Complete either React migration or clearly isolate canvas UI. Establish one input-capture/z-order service, one selected-object store and one panel registry. Remove duplicate modern/legacy build menu and hotbar implementations after parity tests.

P1 Content/data

Move building, weapon, treatment, recipe, event and item definitions toward validated schemas. Add startup validation for duplicate IDs, missing assets, invalid costs, impossible recipes and orphan audio keys.

P2 Memory/GC

Pool bullets, particles, temporary vectors and path nodes; avoid filter/map allocations in hot loops; reuse query arrays; clean event listeners/timers; bound messages and caches; and measure heap growth during multi-day soak tests.

P2 Audio/assets

Lazy-load category manifests, preload only first-session essentials, decode audio on demand with bounded caches, provide format fallbacks, and report missing/duplicate files at build time. Consider image atlases and modern compressed formats.

P2 Build/CI

Add npm scripts and CI gates for type-check, lint, tests, bundle-size budgets, dead-code/import analysis, asset validation and a headless smoke test. Cache npm, use npm ci, pin action versions and publish source maps separately.

P2 Future backend

If saves/cloud/multiplayer are added: use versioned save schemas, delta/compressed snapshots, idempotent APIs, optimistic concurrency, server-side validation, telemetry batching and explicit privacy/retention controls. Keep simulation authority boundaries deliberate.

P3 Developer workflow

Generate the dependency catalog in CI, enforce module boundaries, document ownership, add ADRs, replace .bak source files with Git history, prohibit broad any casts, and create benchmark scenes (100 colonists, dense stockpiles, raid, sandstorm).

11. Concrete Performance Measurement Matrix

Area	Measure	Target / trigger
Frame	p50/p95/p99 frame & update ms	p95 < 16.7 ms at baseline scene
AI	ms by work giver/FSM state; scans/tick	No unbounded full-world scan per colonist
Path	requests, expansions, cache hit, queue age	Bound requests/frame; p95 queue age < 250 ms
Render	draw calls/entities culled/cache hit	Static-layer cache hit > 95% when idle
Memory	heap, allocations/frame, cache sizes	Flat heap over 30-minute soak
Economy	item conservation assertions	Zero negative/duplicated resources
Build	bundle bytes/chunks/build duration	Budget and regression threshold in CI

Appendix A. Complete Script-by-Script Catalog

Each entry is generated from the repository snapshot. “Imports” are direct static/dynamic module specifiers. “Declarations” lists top-level declarations and detected class methods. Checksums make the analyzed snapshot auditable. Runtime call order should be read together with Sections 2–8.

A1. .github/workflows/deploy.yml

Role	GitHub Actions build and Pages deployment pipeline.
Size	38 lines; 807 bytes; SHA-256 prefix 6025b8e67b3f
Direct imports	None
Exports	None detected
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A2. index.html

Role	Browser shell mounting canvas/React roots and main module.
Size	16 lines; 521 bytes; SHA-256 prefix d896e121a4cf
Direct imports	None
Exports	None detected
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A3. package.json

Role	NPM dependency and command manifest.
Size	24 lines; 486 bytes; SHA-256 prefix 03461a0916ed
Direct imports	None
Exports	None detected
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A4. scripts/find_region_refs.sh

Role	Maintenance/search utility; not part of browser runtime.
Size	97 lines; 2812 bytes; SHA-256 prefix 752b986b178f
Direct imports	None
Exports	None detected
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A5. src/assets/audio/manifest.ts

Role	Static asset manifest/loader mapping source assets to runtime keys.
Size	427 lines; 15327 bytes; SHA-256 prefix 2fe1ed236175
Direct imports	None
Exports	AudioVariant, AUDIO_MANIFEST, AudioKey, RandomFn, getAudioVariants, pickAudioVariant, listAudioKeys, resolveAudioSrc
Declarations	fn ensureExtension, fn variants, fn getAudioVariants, fn pickAudioVariant, fn listAudioKeys, fn resolveAudioSrc, type AudioVariant, type VariantInput, type AudioKey, type RandomFn
Detected side effects	reads/writes browser globals; writes diagnostics

A6. src/assets/images.ts

Role	Static asset manifest/loader mapping source assets to runtime keys.
Size	490 lines; 27574 bytes; SHA-256 prefix d13bc3aaacea

Direct imports	./BuildingHouse.png, ./farm/wheat_stage_one.png, ./farm/wheat_stage_two.png, ./farm/wheat_stage_three.png, ./things/colonist/human-like/heads/Male_Average_Normal_east.png, ./things/colonist/human-like/heads/Male_Average_Normal_north.png, ./things/colonist/human-like/heads/Male_Average_Normal_south.png, ./things/colonist/human-like/bodies/Naked_Male_east.png, ./things/colonist/human-like/bodies/Naked_Male_north.png, ./things/colonist/human-like/bodies/Naked_Male_south.png, ./things/colonist/human-like/hair/Afro_east.png, ./things/colonist/human-like/hair/Afro_north.png, ./things/colonist/human-like/hair/Afro_south.png, ./things/colonist/human-like/hair/Bob_east.png, ./things/colonist/human-like/hair/Bob_north.png, ./things/colonist/human-like/hair/Bob_south.png, ./things/colonist/human-like/hair/Bowlcut_east.png, ./things/colonist/human-like/hair/Bowlcut_north.png, ./things/colonist/human-like/hair/Bowlcut_south.png, ./things/colonist/human-like/hair/Braidbun_east.png, 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Exports	ImageAssets
Declarations	class ImageAssets, method getInstance, method loadImage, method getImage, method getColonistSprite, method getAvailableHairStyles, method getAvailableHeadTypes, method getAvailableBodyTypes, method getAvailableApparelTypes, method getWeaponImage, method isLoading
Detected side effects	writes diagnostics

A7. src/core/AdaptiveTickRate.ts

Role	Cross-cutting simulation, performance, pathfinding or rendering utility.
Size	338 lines; 10937 bytes; SHA-256 prefix 8368ef679c4f
Direct imports	None
Exports	ImportanceLevel, ImportanceConfig, IMPORTANCE_CONFIGS, EntityTickState, AdaptiveTickRateManager
Declarations	class AdaptiveTickRateManager, type ImportanceLevel, type ImportanceConfig, type EntityTickState, method constructor, method beginFrame, method shouldUpdate, method forceUpdate, method getEntityUpdateHz, method recordUpdate, method removeEntity, method getStats, method cleanup
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A8. src/core/BudgetedExecution.ts

Role	Cross-cutting simulation, performance, pathfinding or rendering utility.
Size	273 lines; 6928 bytes; SHA-256 prefix 42cb15f9454b
Direct imports	None
Exports	BudgetedTask, BudgetConfig, BudgetedExecutionQueue, BudgetedExecutionManager, budgetedRun
Declarations	class BudgetedExecutionQueue, class BudgetedExecutionManager, fn budgetedRun, type BudgetedTask, type BudgetConfig, method constructor, method enqueue, method remove, method execute, method size, method clear, method getStats, method resetStats, method getInstance, method getQueue, method executeAll, method getAllStats, method resetAllStats
Detected side effects	writes diagnostics

A9. src/core/DirtyRectTracker.ts

Role	Cross-cutting simulation, performance, pathfinding or rendering utility.
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Size	263 lines; 6926 bytes; SHA-256 prefix db8edc76c7cc
Direct imports	None
Exports	DirtyRect, DirtyRectTracker, clearWithDirtyRects
Declarations	class DirtyRectTracker, fn clearWithDirtyRects, type DirtyRect, method constructor, method markFullRedraw, method markDirty, method markCircleDirty, method optimize, method mergeRects, method shouldMerge, method clampRect, method clearDirty, method getDirtyRects, method isDirty, method reset, method resize, method getStats
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A10. src/core/PerformanceMetrics.ts

Role	Cross-cutting simulation, performance, pathfinding or rendering utility.
Size	269 lines; 7475 bytes; SHA-256 prefix 70479f9fc3b0
Direct imports	None
Exports	SubsystemMetrics, LongTask, PerformanceMetrics
Declarations	class PerformanceMetrics, type SubsystemMetrics, type LongTask, method constructor, method getInstance, method startTiming, method endTiming, method recordLongTask, method endFrame, method getAverageMetrics, method getTopOffenders, method getBudgetUtilization, method getSummary, method reset
Detected side effects	writes diagnostics

A11. src/core/RenderCache.ts

Role	Cross-cutting simulation, performance, pathfinding or rendering utility.
Size	389 lines; 11440 bytes; SHA-256 prefix c358b0e1680b
Direct imports	../game/constants, ../game/types, ../game/terrain, ../game/render/index
Exports	WorldBackgroundCache, ColonistSpriteCache, NightOverlayCache, UIPanelCache, ParticleSpriteCache, worldBackgroundCache, colonistSpriteCache, nightOverlayCache, uiPanelCache, particleSpriteCache
Declarations	class WorldBackgroundCache, class ColonistSpriteCache, class NightOverlayCache, class UIPanelCache, class ParticleSpriteCache, method getCanvas, method markDirty, method render, method clear, method size, method clearPanel, method getCacheKey, method getSprite
Detected side effects	reads/writes DOM; writes diagnostics

A12. src/core/RenderOptimizations.ts

Role	Cross-cutting simulation, performance, pathfinding or rendering utility.
Size	298 lines; 7079 bytes; SHA-256 prefix 548e53ec7057
Direct imports	None
Exports	RenderCache, fillRectAlpha, strokeRectAlpha, RenderBatch, GradientCache, drawFastShadow, CanvasStateTracker, renderCache, gradientCache
Declarations	class RenderCache, class RenderBatch, class GradientCache, class CanvasStateTracker, fn fillRectAlpha, fn strokeRectAlpha, fn drawFastShadow, method getCanvas, method clear, method clearAll, method constructor, method add, method execute, method setFillStyle, method setStrokeStyle, method setLineWidth, method setGlobalAlpha, method setFont, method invalidate
Detected side effects	reads/writes DOM

A13. src/core/SimulationClock.ts

Role	Cross-cutting simulation, performance, pathfinding or rendering utility.
Size	220 lines; 5843 bytes; SHA-256 prefix d9d9df41ea6c
Direct imports	None
Exports	SimulationClockConfig, SimulationClock, PositionInterpolator
Declarations	class SimulationClock, class PositionInterpolator, type SimulationClockConfig, method constructor, method getInstance, method tick, method runSimulationStep, method getAlpha, method getFixedDeltaTime, method getFixedDeltaTimeMs, method getSimulationHz, method setSimulationHz, method recordRenderFrame, method getStats, method reset, method interpolateValue, method interpolateRotation
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A14. src/core/particles/impactEffect.ts

Role	Particle creation, simulation and rendering.
Size	145 lines; 4529 bytes; SHA-256 prefix 0d3ae9643d60
Direct imports	../game/types
Exports	createImpactEffect
Declarations	fn createImpactEffect
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A15. src/core/particles/index.ts

Role	Particle creation, simulation and rendering.
Size	2 lines; 29 bytes; SHA-256 prefix fbb026e55453
Direct imports	None
Exports	None detected

Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A16. src/core/particles/muzzleFlash.ts

Role	Particle creation, simulation and rendering.
Size	120 lines; 3974 bytes; SHA-256 prefix ccc60b78f6d6
Direct imports	../game/types
Exports	createMuzzleFlash
Declarations	fn createMuzzleFlash
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A17. src/core/particles/particlePhysics.ts

Role	Particle creation, simulation and rendering.
Size	100 lines; 2760 bytes; SHA-256 prefix b58e7f5ffda7
Direct imports	../game/types
Exports	updateParticles
Declarations	fn updateParticles
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A18. src/core/particles/particleRender.ts

Role	Particle creation, simulation and rendering.
Size	152 lines; 4639 bytes; SHA-256 prefix 8665d3788fc9
Direct imports	../game/types, ../RenderCache
Exports	toggleParticleSprites, isUsingParticleSprites, drawParticles
Declarations	fn toggleParticleSprites, fn isUsingParticleSprites, fn drawParticles, fn renderParticleGroup, fn hexToRgb
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A19. src/core/particles/particles.ts

Role	Particle creation, simulation and rendering.
Size	19 lines; 512 bytes; SHA-256 prefix 514986f1d5e3
Direct imports	None
Exports	createMuzzleFlash, createProjectileTrail, createImpactEffect, createBloodSplatter, createWoodDebris, createStoneDebris, createExplosion, createHealingSparkles, createCraftingSparkles, updateParticles, drawParticles
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A20. src/core/particles/projectileTrail.ts

Role	Particle creation, simulation and rendering.
Size	91 lines; 2882 bytes; SHA-256 prefix a68ee7021364
Direct imports	../game/types
Exports	createProjectileTrail
Declarations	fn createProjectileTrail
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A21. src/core/particles/specialEffects.ts

Role	Particle creation, simulation and rendering.
Size	326 lines; 10478 bytes; SHA-256 prefix 80e66c4f35b7
Direct imports	../game/types
Exports	createBloodSplatter, createWoodDebris, createStoneDebris, createExplosion, createHealingSparkles, createCraftingSparkles
Declarations	fn createBloodSplatter, fn createWoodDebris, fn createStoneDebris, fn createExplosion, fn createHealingSparkles, fn createCraftingSparkles
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A22. src/core/pathfinding.ts

Role	Cross-cutting simulation, performance, pathfinding or rendering utility.
Size	826 lines; 29574 bytes; SHA-256 prefix a118b05b42d7
Direct imports	../game/constants, ../game/terrain

Exports	Grid, makeGrid, clearGrid, clearGridArea, syncTerrainToGrid, syncTerrainToGridPartial, markRectSolid, markCircleSolid, markRectCost, ROAD_COSTS, markRoadPath, updateDirtySections, aStar, computeEnemyPath
Declarations	fn h, fn makeGrid, fn clearGrid, fn clearGridArea, fn syncTerrainToGrid, fn syncTerrainToGridPartial, fn markSectionsDirty, fn markSectionsDirtyByGridCoords, fn markRectSolid, fn markCircleSolid, fn markRectCost, fn isRoadIdx, fn isRoad, fn markRoadPath, fn updateDirtySections, fn clearSection, fn rebuildSection, fn intersectsRect, fn intersectsCircle, fn markRectSolidNoUpdate, fn markCircleSolidNoUpdate, fn markRoadPathNoUpdate, fn tileOf, fn snapRoadCenters, fn makesRightAngleCorner, fn aStar, fn computeEnemyPath, type Grid
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A23. src/core/utils.ts

Role	Cross-cutting simulation, performance, pathfinding or rendering utility.
Size	12 lines; 867 bytes; SHA-256 prefix 4532ced32e93
Direct imports	None
Exports	rand, randi, clamp, Vec2, dist2, length, norm, add, sub, mul, lerp
Declarations	type Vec2
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A24. src/data/itemDatabase.ts

Role	Data catalog queried by gameplay systems.
Size	899 lines; 29108 bytes; SHA-256 prefix 6585d1332074
Direct imports	./game/types, ./game/constants
Exports	ItemDef, getAimTimeSeconds, getCooldownSeconds, ItemDatabase, itemDatabase
Declarations	class ItemDatabase, fn getAimTimeSeconds, fn getCooldownSeconds, type ItemDef, method getInstance, method loadItems, method getItemDef, method getAllItemDefs, method getItemDefsByCategory, method createItem, method getWorkSpeedBonus, method getArmorRating, method getDamage, method getHealAmount, method getNutrition, method getRandomItemsByCategory, method getItemForBackground
Detected side effects	writes diagnostics

A25. src/game/Game.ts

Role	Central composition root and runtime coordinator for input, simulation, managers, rendering and lifecycle.
Size	3266 lines; 138815 bytes; SHA-256 prefix 4612b7e32d43
Direct imports	./core/utils, ./core/pathfinding, ./terrain, ./constants, ./types, ./types/items, ./buildings, ./zones, ./managers/HealthManager, ./managers/InventoryManager, ./managers/ResearchManager, ./ui/panels/researchPanel, ./render, ./managers/WorkGiverManager, ./managers/ReservationManager, ./managers/TaskManager, ./managers/MedicalManager, ./managers/BuildingManager, ./managers/ColonistActionManager, ./managers/ColonistNavigationManager, ./managers/EventManager, ./events, ./enemy_systems/enemyGenerator, ./render/debug/terrainDebugRender, ./colonist_systems/colonistFSM, ./enemy_systems/enemyFSM, ./ui/buildMenu, ./ui/hud/modernHotbar, ./ui/hud/modernBuildMenu, ./ui/hud/controlPanel, ./ui/panels/colonistProfile, ./ui/contextMenu, ./ui/contextMenus/colonistMenu, ./ui/contextMenus/buildings, ./ui/placement, ./ui/mobilePlacement, ./placement/placementSystem, ./colonist_systems/colonistGenerator, ./health/healthSystem, ./audio/helpers/uiAudio, ./health/medicalSystem, ./health/medicalWorkGiver, ./health/surgerySystem, ./core/particles, ./combat/combatSystem, ./combat/pawnCombat, ./managers/CombatManager, ./data/itemDatabase, ./systems/workPriority, ./core/AdaptiveTickRate, ./utils/tileAlignment, ./ui/panels/workPriorityPanel, ./ui/panels/buildingInventoryPanel, ./systems/buildingInventory, ./ui/debugConsole, ./react, ./systems/doorSystem, ./ui/GameOverScreen, ./ui/TutorialSystem, ./systems/treeGrowingZones, ./core/GameState, ./systems/TimeSystem, ./systems/CameraSystem, ./systems/ResourceSystem, ./managers/InputManager, ./managers/UIManager, ./managers/RenderManager, ./managers/NavigationManager, ./core/PerformanceMetrics, ./core/SimulationClock, ./core/BudgetedExecution, ./ui/panels/performanceHUD, ./core/DirtyRectTracker, ./core/RenderCache, ./managers/AudioManager, ./managers/ItemManager, ./managers/ResourceSpawnManager, ./managers/MusicManager, ./events/sandstorm/SandStormSystem, ./ui/dom/mobileControls
Exports	Game
Declarations	class Game, method constructor, method requestFullRebuild, method requestCacheInvalidation, method processQueue, method hasQueuedRebuilds, method getStorageCapacity, method getPopulationCap, method addResource, method getStockpileResourceTotals, method syncResourcesFromStockpiles, method materializeResourcesToStockpiles, method consumeStockpileResource, method consumeStockpileCost, method releaseColonistReservations, method cleanupMiningAssignments, method getMoveSpeedMultiplier, method getWorkSpeedMultiplier, method getArmorReduction, method tryConsumeInventoryFood, method recalcInventoryWeight, method clampCameraToWorld, method calculateUIScale, method getMobileControlsInstance, method syncMobileControls, method getMobileControls, method applyTouchUIState, method setTouchUIEnabled, method refreshTouchUIState, method getScaledFont, method scale, method getZoneDragPreviewRect, method updateZoomOverlay, method bindInput, method handleTapOrClickAtScreen, method findColonistAt, method findEnemyAt, method findBuildingAt, method msg, method queueOperation, method cancelOperation, method calculateOperationSuccessChance, method toast, method playAudio, method stopAudio, method updateMusic, method scatter, method tryRespawn, method buildHQ, method newGame, method spawnEnemy, method placeAtMouse, method tryPlaceNow, method nudgePending, method rotatePending, method confirmPending, method cancelPending, method paintPathAtMouse, method paintWallAtMouse, method cancelOrErase, method evictColonistsFrom, method buildingCapacity, method buildingHasSpace, method reserveSleepSpot, method releaseSleepReservation, method tryEnterBuilding, method leaveBuilding, method releaseBuildReservation, method clearPath, method pickTask, method isProtectedByTurret, method centerOf, method updateTurret, method isNight, method spawnWave, method nextDay, method dayTick, method keyPressed, method update, method draw
Detected side effects	registers event listeners; schedules animation frames; schedules timers; reads/writes DOM; reads/writes browser globals; writes diagnostics

A26. src/game/audio/buildingAudioMap.ts

Role	Supporting project script or configuration surface.
Size	389 lines; 12117 bytes; SHA-256 prefix 42eb53d40ac3
Direct imports	./managers/AudioManager, ./types
Exports	AudioClip, AudioVariant, BuildingAudioConfig, getBuildingAudio, normalizeAudioVariant, pickRandomVariant, getConstructionAudio, getConstructionCompleteAudio, getBuildingPlacementAudio, getOperationalAudio
Declarations	fn getBuildingAudio, fn normalizeAudioVariant, fn pickRandomVariant, fn getConstructionAudio, fn getConstructionCompleteAudio, fn getBuildingPlacementAudio, fn getOperationalAudio, type AudioClip, type AudioVariant, type BuildingAudioConfig
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A27. src/game/audio/helpers/constructionAudio.ts

Role	Supporting project script or configuration surface.
Size	14 lines; 466 bytes; SHA-256 prefix 81b6aee16634

Direct imports	None
Exports	stopConstructionLoop, stopConstructionAudio
Declarations	fn stopConstructionLoop, fn stopConstructionAudio
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A28. src/game/audio/helpers/medicalAudio.ts

Role	Supporting project script or configuration surface.
Size	73 lines; 2184 bytes; SHA-256 prefix be7bc9fb346
Direct imports	../managers/AudioManager
Exports	stopMedicalAudio, startTendingAudio, startSurgeryAudio
Declarations	fn getMedicalAudioState, fn stopActiveAudio, fn stopMedicalAudio, fn startTendingAudio, fn startSurgeryAudio, type MedicalAudioState
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A29. src/game/audio/helpers/musicAudio.ts

Role	Supporting project script or configuration surface.
Size	161 lines; 4970 bytes; SHA-256 prefix 550e605dca26
Direct imports	../managers/AudioManager, ../assets/audio/manifest
Exports	playRaidMusic, stopRaidMusic, playDayMusic, stopDayMusic, playGameOverMusic, stopGameOverMusic
Declarations	fn safePlayLoop, fn getShuffleState, fn getShuffleSession, fn shuffleOrder, fn nextShuffledIndex, fn rngForVariantIndex, fn playShuffledLoop, fn stopShuffledLoop, fn playRaidMusic, fn stopRaidMusic, fn playDayMusic, fn stopDayMusic, fn playGameOverMusic, fn stopGameOverMusic, type MusicKey, type ShuffleState, type ShuffleSession
Detected side effects	registers event listeners; writes diagnostics

A30. src/game/audio/helpers/uiAudio.ts

Role	Supporting project script or configuration surface.
Size	44 lines; 966 bytes; SHA-256 prefix 55af810cf22d
Direct imports	../managers/AudioManager
Exports	playUiClickPrimary, playUiClickSecondary, playUiPanelOpen, playUiPanelClose, playUiHover, playUiHotbarHover, playUiDragStart, playUiDragEnd
Declarations	fn playUi, fn playUiClickPrimary, fn playUiClickSecondary, fn playUiPanelOpen, fn playUiPanelClose, fn playUiHover, fn playUiHotbarHover, fn playUiDragStart, fn playUiDragEnd
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A31. src/game/audio/weaponAudioMap.ts

Role	Supporting project script or configuration surface.
Size	205 lines; 7139 bytes; SHA-256 prefix adbf29774125
Direct imports	../managers/AudioManager
Exports	WeaponAudioContext, getWeaponFireAudio, getMeleelImpactAudio, getWeaponAudio, getWeaponAudioContext, getWeaponAudioByDefName
Declarations	fn getWeaponFireAudio, fn getMeleelImpactAudio, fn getWeaponAudio, fn getWeaponAudioContext, fn getWeaponAudioByDefName, type WeaponAudioContext
Detected side effects	uses Web Audio

A32. src/game/buildings.ts

Role	Supporting project script or configuration surface.
Size	337 lines; 10365 bytes; SHA-256 prefix 841bc6e83bc5
Direct imports	./constants, ./types, ./systems/buildingInventory, ./zones
Exports	BUILD_TYPES, hasCost, payCost, refundCost, makeBuilding, costText, groupByCategory
Declarations	fn hasCost, fn payCost, fn refundCost, fn makeBuilding, fn costText, fn groupByCategory
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A33. src/game/colonist_systems/colonistData.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	30 lines; 1061 bytes; SHA-256 prefix df93a0f968d4
Direct imports	None
Exports	NAMES, FAVORITE_FOODS, randomChoice
Declarations	fn randomChoice
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A34. src/game/colonist_systems/colonistFSM.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	2297 lines; 93863 bytes; SHA-256 prefix 5cac96a0a522
Direct imports	../core/utlis, ../constants, ../types, ../types/items, /skills, ../health/healthSystem, ../health/medicalSystem, ../health/medicalWorkGiver, ../health/surgerySystem, ../audio/helpers/medicalAudio, ../audio/helpers/constructionAudio, ../systems/doorSystem, ../data/itemDatabase, ../audio/buildingAudioMap, ../buildings, ../terrain, /states, ../systems/colonistIntent, ../core/particles
Exports	updateColonistFSM
Declarations	fn clampToWorld, fn ensureColonistMovementTracking, fn ensureColonistInventory, fn findClosestRestBed, fn resetDoorWaitState, fn wouldCollideWithBuildings, fn getSpeedMultiplier, fn moveTowardsSafely, fn checkAndRescueStuckColonist, fn updateColonistFSM, fn changeState, fn evaluateIntent
Detected side effects	writes diagnostics

A35. src/game/colonist_systems/colonistGenerator.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	1012 lines; 42875 bytes; SHA-256 prefix 95f0763c8351
Direct imports	../assets/images, ../data/itemDatabase, /colonistData, /traits, /traits/passiveTraits, /traits/backgrounds, /narrative, /skills, ../types
Exports	ColonistProfile, generateColonistProfile, getColonistDescription, getColonistMood
Declarations	class SeededRandom, fn hashStringToNumber, fn applyBackstorySkillBoosts, fn generateStats, fn generateContextualTraits, fn generateDetailedInfo, fn getSkillsForBackground, fn createRichBackstory, fn generateStartingInventory, fn getEquipSlot, fn getRandomQuality, fn generateColonistProfile, fn getColonistDescription, fn getColonistMood, type ColonistProfile, method constructor, method next, method range
Detected side effects	writes diagnostics

A36. src/game/colonist_systems/index.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	6 lines; 155 bytes; SHA-256 prefix adbc30687c29
Direct imports	None
Exports	None detected
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A37. src/game/colonist_systems/narrative/birthplaces.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	31 lines; 866 bytes; SHA-256 prefix dff0b69b3793
Direct imports	None
Exports	BIRTHPLACES
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A38. src/game/colonist_systems/narrative/fears.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	14 lines; 541 bytes; SHA-256 prefix 919de3ad72c3
Direct imports	None
Exports	FEARS
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A39. src/game/colonist_systems/narrative/index.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	10 lines; 288 bytes; SHA-256 prefix f3047313e937
Direct imports	None
Exports	BIRTHPLACES, LIFE_EVENTS, type LifeEvent, SKILLS, FEARS, SECRETS
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A40. src/game/colonist_systems/narrative/lifeEvents.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	54 lines; 3590 bytes; SHA-256 prefix 401792bff5cd
Direct imports	None
Exports	LifeEvent, LIFE_EVENTS
Declarations	type LifeEvent

Detected side effects	none detected by static scan; behavior is exposed through imports/exports
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A41. src/game/colonist_systems/narrative/secrets.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	42 lines; 1735 bytes; SHA-256 prefix 4d735edba521
Direct imports	None
Exports	SECRETS
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A42. src/game/colonist_systems/narrative/skills.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	15 lines; 586 bytes; SHA-256 prefix a4cf3430765a
Direct imports	None
Exports	SKILLS
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A43. src/game/colonist_systems/skills.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	97 lines; 3606 bytes; SHA-256 prefix 09850ffc906d
Direct imports	../types
Exports	createDefaultSkillSet, addStartingSkillVariance, getSkill, grantSkillXP, skillLevel, skillWorkSpeedMultiplier, skillSuccessBonus, medicineSuccessChance
Declarations	fn createDefaultSkillSet, fn addStartingSkillVariance, fn getSkill, fn grantSkillXP, fn skillLevel, fn skillWorkSpeedMultiplier, fn skillSuccessBonus, fn medicineSuccessChance
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A44. src/game/colonist_systems/states/buildState.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	266 lines; 10811 bytes; SHA-256 prefix 911766d89706
Direct imports	../types, ../Game, ../constants, ../buildings, ../audio/buildingAudioMap, ../audio/helpers/constructionAudio, ../systems/doorSystem, ../terrain, ../skills
Exports	updateBuildState
Declarations	fn wouldCollideWithBuildings, fn updateBuildState
Detected side effects	writes diagnostics

A45. src/game/colonist_systems/states/cookingState.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	199 lines; 5814 bytes; SHA-256 prefix a1f289cf6dd3
Direct imports	../types, ../Game, ../skills
Exports	updateCookingState
Declarations	fn updateCookingState, fn handleWheatPickup, fn handleCooking, fn completeCooking, fn abandonCooking
Detected side effects	writes diagnostics

A46. src/game/colonist_systems/states/coolingState.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	229 lines; 6995 bytes; SHA-256 prefix e784793d5852
Direct imports	../types, ../Game, ../skills, ../types/items
Exports	updateCoolingState
Declarations	fn updateCoolingState, fn handleHottingotPickup, fn handleCooling, fn completeCooling, fn abandonCooling
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A47. src/game/colonist_systems/states/equipmentState.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	195 lines; 5672 bytes; SHA-256 prefix 23d316688e4b
Direct imports	../types, ../Game, ../types/items, ../data/itemDatabase
Exports	updateEquipmentState, findBestWeaponUpgrade

Declarations	fn updateEquipmentState, fn handleWeaponPickupAndEquip, fn isWeapon, fn abandonEquipment, fn findBestWeaponUpgrade, fn getWeaponValue
Detected side effects	writes diagnostics

A48. src/game/colonist_systems/states/guardState.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	39 lines; 1083 bytes; SHA-256 prefix c9d4cff285b
Direct imports	../types, ../Game
Exports	updateGuardState
Declarations	fn updateGuardState
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A49. src/game/colonist_systems/states/idleState.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	94 lines; 3096 bytes; SHA-256 prefix 8ae31e169353
Direct imports	../types, ../Game
Exports	updateIdleState
Declarations	fn updateIdleState
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A50. src/game/colonist_systems/states/index.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	22 lines; 965 bytes; SHA-256 prefix 1967bc67780f
Direct imports	None
Exports	updateCookingState, updateStonecuttingState, updateSmeltingState, updateCoolingState, updateSmithingState, updateEquipmentState, findBestWeaponUpgrade, updateMineState, updateRestingState, updateSleepState, updateGoToSleepState, updateResearchState, updateIdleState, updateMoveState, updateGuardState, updateBuildState
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A51. src/game/colonist_systems/states/mineState.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	237 lines; 9443 bytes; SHA-256 prefix d7ca3b6d7725
Direct imports	../types, ../Game, ../constants, ../terrain, ../skills, ../types/items, ../core/particles
Exports	updateMineState
Declarations	fn updateMineState
Detected side effects	writes diagnostics

A52. src/game/colonist_systems/states/moveState.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	100 lines; 3177 bytes; SHA-256 prefix bf89096e2411
Direct imports	../types, ../Game
Exports	updateMoveState
Declarations	fn updateMoveState
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A53. src/game/colonist_systems/states/researchState.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	55 lines; 1674 bytes; SHA-256 prefix 60e9557f9d27
Direct imports	../types, ../Game
Exports	updateResearchState
Declarations	fn updateResearchState
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A54. src/game/colonist_systems/states/restingSleepState.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	191 lines; 7077 bytes; SHA-256 prefix a78682eb90d5
Direct imports	../types, ../Game

Exports	updateRestingState, updateSleepState, updateGoToSleepState
Declarations	fn updateRestingState, fn updateSleepState, fn updateGoToSleepState
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A55. src/game/colonist_systems/states/smeltingState.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	268 lines; 8353 bytes; SHA-256 prefix 01db03bf2dbe
Direct imports	../types, ../Game, ../types/items, ../skills
Exports	updateSmeltingState
Declarations	fn updateSmeltingState, fn handleOrePickup, fn handleSmelting, fn completeSmelting, fn abandonSmelting
Detected side effects	writes diagnostics

A56. src/game/colonist_systems/states/smithingState.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	335 lines; 10510 bytes; SHA-256 prefix ebb098affa04
Direct imports	../types, ../Game, ../types/items, ../skills
Exports	updateSmithingState
Declarations	fn updateSmithingState, fn determineRecipeToCraft, fn hasRequiredMaterials, fn handleMaterialCollection, fn collectMaterial, fn handleSmithing, fn completeSmithing, fn abandonSmithing, type SmithingRecipe
Detected side effects	writes diagnostics

A57. src/game/colonist_systems/states/stonecuttingState.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	198 lines; 6062 bytes; SHA-256 prefix 861e2fe65684
Direct imports	../types, ../Game, ../skills
Exports	updateStonecuttingState
Declarations	fn updateStonecuttingState, fn handleRubblePickup, fn handleStonecutting, fn completeStonecutting, fn abandonStonecutting
Detected side effects	writes diagnostics

A58. src/game/colonist_systems/traits/appearance.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	159 lines; 6317 bytes; SHA-256 prefix 4610cfdc7e54
Direct imports	None
Exports	ColorVariant, AppearanceTraits, HAIR_COLORS, SKIN_TONES, EYE_COLORS, HAIR_STYLES, BODY_TYPES, CLOTHING_COLORS, getRandomColor, getRandomHairStyle, getRandomBodyType, generateAppearanceTraits, getColorTint
Declarations	fn getRandomColor, fn getRandomHairStyle, fn getRandomBodyType, fn generateAppearanceTraits, fn getColorTint, type ColorVariant, type AppearanceTraits
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A59. src/game/colonist_systems/traits/backgrounds.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	230 lines; 6071 bytes; SHA-256 prefix 6c10a3d6f40d
Direct imports	None
Exports	Background, BACKGROUNDS, ORIGIN_STORIES, getRandomBackground, getRandomOriginStory
Declarations	fn getRandomBackground, fn getRandomOriginStory, type Background
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A60. src/game/colonist_systems/traits/index.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	37 lines; 1164 bytes; SHA-256 prefix 28152b461de6
Direct imports	./backgrounds, ./appearance, ./passiveTraits
Exports	ColonistTraits, generateCompleteTraitSet
Declarations	fn generateCompleteTraitSet, type ColonistTraits
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A61. src/game/colonist_systems/traits/passiveTraits.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	496 lines; 16272 bytes; SHA-256 prefix 7c0a97aaa6a0

Direct imports	None
Exports	PassiveTrait, TraitEffect, PHYSICAL_TRAITS, MENTAL_TRAITS, SOCIAL_TRAITS, SPECIAL_TRAITS, ALL_PASSIVE_TRAITS, getRandomPassiveTrait, generatePassiveTraits, applyTraitEffects, getTraitsByCategory, getTraitById
Declarations	fn getRandomPassiveTrait, fn generatePassiveTraits, fn applyTraitEffects, fn getTraitsByCategory, fn getTraitById, type PassiveTrait, type TraitEffect
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A62. src/game/combat/combatSystem.ts

Role	Combat calculations, targeting, projectiles or pawn attacks.
Size	401 lines; 15564 bytes; SHA-256 prefix 9a66efa9368d
Direct imports	../Game, ../types, ../colonist_systems/skills, ../../core/particles, ../../core/utlis, ../audio/weaponAudioMap, ../../data/itemDatabase
Exports	updateTurret, updateProjectiles
Declarations	fn lineIntersectsRect, fn hasLineOfFire, fn pickTarget, fn updateTurret, fn fireTurretShot, fn updateProjectiles
Detected side effects	writes diagnostics

A63. src/game/combat/pawnCombat.ts

Role	Combat calculations, targeting, projectiles or pawn attacks.
Size	801 lines; 31773 bytes; SHA-256 prefix 0dd675a5305b
Direct imports	../Game, ../types, ../colonist_systems/skills, ../../data/itemDatabase, ../../core/particles, ../audio/weaponAudioMap, ./unarmedCombat
Exports	updateColonistCombat
Declarations	fn lineIntersectsRect, fn hasLineOfFire, fn segmentIntersectsCircle, fn willHitFriendly, fn getDirectionalCoverMultiplier, fn interpolateAccuracy, fn getDistanceCoverMultiplier, fn coverPenalty, fn getWeaponStats, fn pickTarget, fn updateColonistCombat, type CoverSource
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A64. src/game/combat/unarmedCombat.ts

Role	Combat calculations, targeting, projectiles or pawn attacks.
Size	141 lines; 3908 bytes; SHA-256 prefix 19a89dc0496
Direct imports	None
Exports	UnarmedAttack, UNARMED_ATTACKS, selectUnarmedAttack, calculateUnarmedDamage, getUnarmedAverageDPS, isUnarmed
Declarations	fn selectUnarmedAttack, fn calculateUnarmedDamage, fn getUnarmedAverageDPS, fn isUnarmed, type UnarmedAttack
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A65. src/game/combat/weaponStats.ts

Role	Combat calculations, targeting, projectiles or pawn attacks.
Size	88 lines; 2987 bytes; SHA-256 prefix f122ed040501
Direct imports	../../data/itemDatabase, ../constants
Exports	calculateOptimalDPS, getAccuracyAtRange, calculateDPSAtRange, getWeaponStatsSummary
Declarations	fn calculateOptimalDPS, fn getAccuracyAtRange, fn calculateDPSAtRange, fn getWeaponStatsSummary
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A66. src/game/constants.ts

Role	Supporting project script or configuration surface.
Size	37 lines; 1430 bytes; SHA-256 prefix 038a9302aff8
Direct imports	./types
Exports	T, WORLD, HQ_POS, COLORS, NIGHT_SPAN, RIMWORLD_TICKS_PER_SECOND, GAME_TICKS_PER_SECOND, rimworldTicksToSeconds, secondsToTicks
Declarations	fn rimworldTicksToSeconds, fn secondsToTicks
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A67. src/game/core/GameState.ts

Role	Cross-cutting simulation, performance, pathfinding or rendering utility.
Size	148 lines; 3238 bytes; SHA-256 prefix 5277a4bf652d
Direct imports	../types, ../buildings
Exports	Resources, TreeEntity, RockEntity, GameState
Declarations	class GameState, type Resources, type TreeEntity, type RockEntity, method isNight, method reset
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A68. src/game/enemy_systems/enemyFSM.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
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Size	512 lines; 18120 bytes; SHA-256 prefix f5cf7a71614a
Direct imports	../core/utlis, ../constants, ../types, ../systems/doorSystem, ../core/pathfinding, ../terrain, ../data/itemDatabase, ../combat/unarmedCombat, ../audio/weaponAudioMap
Exports	updateEnemyFSM
Declarations	fn wouldCollideWithBuildings, fn getEnemyPathStartIndex, fn ensureEnemyPath, fn moveEnemyAlongPath, fn moveDirectlyWithCollision, fn updateEnemyFSM
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A69. src/game/enemy_systems/enemyGenerator.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	310 lines; 9209 bytes; SHA-256 prefix b6a66647c234
Direct imports	../assets/images, ../data/itemDatabase, ../colonist_systems/colonistData, ../colonist_systems/traits, ../types
Exports	EnemyRole, EnemyProfile, generateEnemyProfile, createEnemyWithProfile, getEnemyDescription
Declarations	fn generateEnemyProfile, fn selectWeaponForEnemy, fn calculateEnemyStats, fn createEnemyWithProfile, fn getEnemyDescription, type EnemyRole, type EnemyProfile
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A70. src/game/events/CacheWindfallEvent.ts

Role	Incident/event definitions and event lifecycle behavior.
Size	30 lines; 1007 bytes; SHA-256 prefix d0b48cc7859c
Direct imports	./types, ./eventUtils
Exports	CacheWindfallEvent
Declarations	class CacheWindfallEvent, method canTrigger, method trigger
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A71. src/game/events/CampfireStoriesEvent.ts

Role	Incident/event definitions and event lifecycle behavior.
Size	33 lines; 1078 bytes; SHA-256 prefix 01d84f6b3d84
Direct imports	./types, ./eventUtils
Exports	CampfireStoriesEvent
Declarations	class CampfireStoriesEvent, method canTrigger, method trigger
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A72. src/game/events/ColonyIncidents.ts

Role	Incident/event definitions and event lifecycle behavior.
Size	6 lines; 287 bytes; SHA-256 prefix 7ea28d7cfb60
Direct imports	None
Exports	SolarFlareEvent, FoodSpoilageEvent, CacheWindfallEvent, CampfireStoriesEvent, SkyFestivalEvent
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A73. src/game/events/FoodSpoilageEvent.ts

Role	Incident/event definitions and event lifecycle behavior.
Size	36 lines; 1260 bytes; SHA-256 prefix a96c737c5ba0
Direct imports	./types, ./eventUtils
Exports	FoodSpoilageEvent
Declarations	class FoodSpoilageEvent, method canTrigger, method trigger
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A74. src/game/events/SingleEnemyAttackEvent.ts

Role	Incident/event definitions and event lifecycle behavior.
Size	30 lines; 940 bytes; SHA-256 prefix d509b7b22f5a
Direct imports	./types
Exports	SingleEnemyAttackEvent
Declarations	class SingleEnemyAttackEvent, method canTrigger, method trigger
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A75. src/game/events/SkyFestivalEvent.ts

Role	Incident/event definitions and event lifecycle behavior.
Size	35 lines; 1083 bytes; SHA-256 prefix 7726ef3bd248
Direct imports	./types, ./eventUtils
Exports	SkyFestivalEvent
Declarations	class SkyFestivalEvent, method canTrigger, method trigger
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A76. src/game/events/SolarFlareEvent.ts

Role	Incident/event definitions and event lifecycle behavior.
Size	32 lines; 1058 bytes; SHA-256 prefix a4537cd72026
Direct imports	./types, ./eventUtils
Exports	SolarFlareEvent
Declarations	class SolarFlareEvent, method canTrigger, method trigger
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A77. src/game/events/eventUtils.ts

Role	Incident/event definitions and event lifecycle behavior.
Size	33 lines; 1143 bytes; SHA-256 prefix 535bccb4f18f
Direct imports	../constants, ../types/items
Exports	isGameReadyForEvents, livingColonists, pickColonist, dropAroundHQ
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A78. src/game/events/index.ts

Role	Incident/event definitions and event lifecycle behavior.
Size	21 lines; 734 bytes; SHA-256 prefix 45b2a5e46d87
Direct imports	./types, ./CacheWindfallEvent, ./CampfireStoriesEvent, ./FoodSpoilageEvent, ./SingleEnemyAttackEvent, ./SkyFestivalEvent, ./SolarFlareEvent, ./sandstorm/SandStormEvent
Exports	createDefaultEvents
Declarations	fn createDefaultEvents
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A79. src/game/events/sandstorm/SandStormEvent.ts

Role	Incident/event definitions and event lifecycle behavior.
Size	31 lines; 1006 bytes; SHA-256 prefix aba357025acc
Direct imports	./types, ./SandStormSystem
Exports	SandStormEvent
Declarations	class SandStormEvent, method canTrigger, method trigger
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A80. src/game/events/sandstorm/SandStormSystem.ts

Role	Incident/event definitions and event lifecycle behavior.
Size	140 lines; 4121 bytes; SHA-256 prefix 9676adc29600
Direct imports	../constants, ../Game, ../types
Exports	SandStormSystem
Declarations	class SandStormSystem, method constructor, method isActive, method getVisionRadius, method start, method update, method getVisionSources, method isPointVisible, method renderFog, method lineIntersectsRect
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A81. src/game/events/types.ts

Role	Incident/event definitions and event lifecycle behavior.
Size	23 lines; 468 bytes; SHA-256 prefix ac8167452614
Direct imports	./Game
Exports	GameEventCategory, EventContext, GameEvent
Declarations	type GameEventCategory, type EventContext, type GameEvent
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A82. src/game/health/healthSystem.ts

Role	Health, treatment, medical work and surgery domain logic.
Size	713 lines; 24977 bytes; SHA-256 prefix 519c4cf5f797
Direct imports	./types
Exports	DEFAULT_BODY_PARTS, initializeColonistHealth, calculateOverallHealth, updateHealthStats, selectRandomBodyPart, createInjury, damageBodyPart, DamageOptions, applyDamageToColonist, handleColonistDeath, tickBleeding, tickInfections, updateHealthProgression, getHealthStatus, healInjuries, getInjurySummary, getMostCriticalInjury, basicFieldTreatment, calculateImmunityGainRate, calculateHealingPointsPerDay
Declarations	fn initializeColonistHealth, fn calculateOverallHealth, fn updateHealthStats, fn selectRandomBodyPart, fn createInjury, fn damageBodyPart, fn applyDamageToColonist, fn handleColonistDeath, fn tickBleeding, fn tickInfections, fn updateHealthProgression, fn getHealthStatus, fn healInjuries, fn getInjurySummary, fn getMostCriticalInjury, fn getInjuryBaseValues, fn basicFieldTreatment, fn calculateImmunityGainRate, fn calculateHealingPointsPerDay, type DamageOptions
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A83. src/game/health/medicalSystem.ts

Role	Health, treatment, medical work and surgery domain logic.
Size	289 lines; 10315 bytes; SHA-256 prefix 48463fa2894c
Direct imports	./types, ./colonist_systems/skills, ./healthSystem, .././data/itemDatabase, ./medicalWorkGiver
Exports	MedicalTreatment, MedicalPriority, MEDICAL_TREATMENTS, MedicalSystem, medicalSystem
Declarations	class MedicalSystem, type MedicalTreatment, type MedicalPriority, method performTreatment, method applySuccessfulTreatment, method applyFailedTreatment, method hasRequiredItems, method consumelItems, method getDoctorSkill
Detected side effects	writes diagnostics

A84. src/game/health/medicalWorkGiver.ts

Role	Health, treatment, medical work and surgery domain logic.
Size	728 lines; 23511 bytes; SHA-256 prefix e80c03a93036
Direct imports	./types, ./medicalSystem, ./colonist_systems/skills
Exports	MedicalJob, MedicalWorkGiver, medicalWorkGiver
Declarations	class MedicalWorkGiver, type MedicalJob, method scanForMedicalWork, method scanForSurgeryWork, method scanForFeedingWork, method scanEmergencyMedical, method scanNormalMedical, method findBestTreatment, method findTargetInjury, method findUntreatedInjury, method calculatePatientUrgency, method calculatePatientPriority, method calculateTreatmentPriority, method getReservedJob, method findPlayerForcedJob, method reserveJob, method releaseJob, method completeJob, method isPatientReserved, method cleanupJobs, method getDoctorSkill, method getColonistId, method getActiveJobs
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A85. src/game/health/surgerySystem.ts

Role	Health, treatment, medical work and surgery domain logic.
Size	764 lines; 22173 bytes; SHA-256 prefix 451d797f3915
Direct imports	./types, ./healthSystem
Exports	executeSurgery, calculateSurgerySuccessChance, getHospitalBedBonus, getCleanlinessBonus, getOperationComplexity
Declarations	fn createSurgeryInjury, fn updatePatientHealth, fn executeSurgery, fn calculateSurgerySuccessChance, fn getHospitalBedBonus, fn getCleanlinessBonus, fn getOperationComplexity, fn performSurgeryOperation, fn performAmputation, fn performProstheticInstallation, fn performImplantInstallation, fn performOrganTransplant, fn performOrganHarvest, fn performScarRemoval, fn performInfectionTreatment, fn handleSurgeryFailure, fn getNearbyBodyParts, fn getProstheticEfficiency, fn getProstheticLabel, fn getImplantEfficiencyBonus, fn getImplantLabel
Detected side effects	writes diagnostics

A86. src/game/managers/AudioManager.ts

Role	Manager boundary coordinating a focused domain against Game/state.
Size	408 lines; 14451 bytes; SHA-256 prefix d0784d9d36d5
Direct imports	.././assets/audio/manifest, .././core/utlis
Exports	PlayAudioOptions, AudioManager
Declarations	class AudioManager, fn deriveCategory, type PlayAudioOptions, type CachedBuffer, type ActiveSound, method getInstance, method constructor, method initAudioContext, method preload, method play, method stop, method stopAll, method setMasterVolume, method getMasterVolume, method setCategoryVolume, method getCategoryVolume, method setMuted, method isMuted, method setListenerPosition, method getListenerPosition, method getListenerZoom, method loadBuffer, method computeVolume, method refreshSoundVolumes, method calculateCameraAltitude, method updateSpatialAudio
Detected side effects	registers event listeners; reads/writes browser globals; creates audio elements; uses Web Audio; writes diagnostics

A87. src/game/managers/BuildingManager.ts

Role	Manager boundary coordinating a focused domain against Game/state.
Size	166 lines; 5144 bytes; SHA-256 prefix de5e65da10d3
Direct imports	./types, ../Game, .././core/utlis
Exports	BuildingManager
Declarations	class BuildingManager, method constructor, method centerOf, method buildingHasSpace, method isProtectedByTurret, method findCompletedBuildings, method findNearestBuilding, method findAvailableBuildings, method tryEnterBuilding, method getStorageBuildings, method getMedicalBuildings
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A88. src/game/managers/ColonistActionManager.ts

Role	Manager boundary coordinating a focused domain against Game/state.
Size	72 lines; 2191 bytes; SHA-256 prefix da86329df1f3
Direct imports	../types, ../Game, ../ui/contextMenu
Exports	ColonistActionManager
Declarations	class ColonistActionManager, method constructor, method handleContextMenuAction, method handleDraftToggle
Detected side effects	writes diagnostics

A89. src/game/managers/ColonistMovementManager.ts

Role	Manager boundary coordinating a focused domain against Game/state.
Size	1 lines; 0 bytes; SHA-256 prefix e3b0c44298fc
Direct imports	None
Exports	None detected
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A90. src/game/managers/ColonistNavigationManager.ts

Role	Manager boundary coordinating a focused domain against Game/state.
Size	570 lines; 21679 bytes; SHA-256 prefix a50f0e303d6b
Direct imports	../types, ../Game, ../constants, ../systems/doorSystem
Exports	ColonistNavigationManager
Declarations	class ColonistNavigationManager, method constructor, method forceColonistToRest, method forceColonistToEat, method sendColonistToHQ, method sendColonistToSafety, method sendColonistToBed, method sendColonistToFood, method sendColonistToBuilding, method sendColonistToNearestBuilding, method sendColonistToPosition, method makeColonistGuard, method evacuateAllColonists, method clearPath, method getMoveSpeedMultiplier, method enableNewPathfinding, method disableNewPathfinding, method isUsingNewPathfinding
Detected side effects	writes diagnostics

A91. src/game/managers/ColonistTaskManager.ts

Role	Manager boundary coordinating a focused domain against Game/state.
Size	1 lines; 0 bytes; SHA-256 prefix e3b0c44298fc
Direct imports	None
Exports	None detected
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A92. src/game/managers/CombatManager.ts

Role	Manager boundary coordinating a focused domain against Game/state.
Size	942 lines; 32108 bytes; SHA-256 prefix 5f329b4d279c
Direct imports	../Game, ../types, ../colonist_systems/skills, ../data/itemDatabase
Exports	ThreatAssessment, CombatPosition, CombatAssignment, CombatSituation, CombatManager
Declarations	class CombatManager, type ThreatAssessment, type CombatPosition, type CombatAssignment, type CombatSituation, type CoverSource, method constructor, method evaluateCombatSituation, method getCombatAssignment, method shouldFlee, method getBestTarget, method findCoverPosition, method shouldTakeCover, method getDangerState, method calculateSituation, method assessThreats, method calculateRetreatDecision, method calculateCoverPositions, method getPositionsAroundWall, method getPositionsAroundCircle, method getDirectionalCoverMultiplier, method getDistanceCoverMultiplier, method getCoverValueAtPosition, method findSafePosition, method assignCombatRole, method createRetreatAssignment, method isCriticallyInjured, method hasRangedWeapon, method getWeaponRange, method lineIntersectsRect, method findNearestEnemy, method getColonistId, method cleanup
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A93. src/game/managers/EventManager.ts

Role	Manager boundary coordinating a focused domain against Game/state.
Size	124 lines; 3423 bytes; SHA-256 prefix 2c7ffc6fd622
Direct imports	../Game, ../events/types
Exports	EventManagerOptions, EventManager
Declarations	class EventManager, type EventManagerOptions, method constructor, method setEnabled, method isEnabled, method getEvents, method registerEvent, method update, method triggerEventById, method triggerRandomEligibleEvent, method tryTriggerRandomEvent, method pickWeightedEligibleEvent, method canTriggerEvent, method buildContext
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A94. src/game/managers/HaulManager.ts

Role	Manager boundary coordinating a focused domain against Game/state.
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Size	225 lines; 7134 bytes; SHA-256 prefix a97ce66271a4
Direct imports	../core/utlis, ../types/items, ../types/stockpiles
Exports	HaulingJob, ConstructionMaterialRequest, HaulManager
Declarations	class HaulManager, type HaulingJob, type ConstructionMaterialRequest, method createHaulingJob, method createConstructionRequest, method createMaterialDeliveryJobs, method calculateHaulingPriority, method getNextHaulingJob, method calculateDistance, method completeJob, method failJob, method cancelJob, method getJobsForColonist, method getAvailableJobs, method getMaterialRequests, method isConstructionReady, method deliverMaterialToConstruction, method removeConstructionRequest, method cleanupOldJobs
Detected side effects	schedules timers

A95. src/game/managers/HealthManager.ts

Role	Manager boundary coordinating a focused domain against Game/state.
Size	263 lines; 8753 bytes; SHA-256 prefix 5ef71132cc53
Direct imports	../types
Exports	HealthManager
Declarations	class HealthManager, method calculatePainFromDamage, method calculateBleedingFromDamage, method calculateHealRate, method calculateInfectionChance, method generateInjuryDescription, method recalculateColonistHealth, method getHealthSummary
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A96. src/game/managers/InputManager.ts

Role	Manager boundary coordinating a focused domain against Game/state.
Size	205 lines; 4943 bytes; SHA-256 prefix c06aa9b378fe
Direct imports	None
Exports	MouseState, InputManager
Declarations	class InputManager, type MouseState, method bindKeyboardEvents, method setMousePosition, method setMouseDown, method setMouseRightDown, method getMouse, method getMouseRef, method setKeyState, method isKeyPressed, method keyPressed, method getKeyStateRef, method getTouchPan, method setTouchDist, method getTouchDist, method setLastInputWasTouch, method wasLastInputTouch, method reset
Detected side effects	registers event listeners; reads/writes browser globals

A97. src/game/managers/InventoryManager.ts

Role	Manager boundary coordinating a focused domain against Game/state.
Size	307 lines; 9958 bytes; SHA-256 prefix bdf6252a9a2c
Direct imports	../types, ../data/itemDatabase
Exports	InventoryManager
Declarations	class InventoryManager, method getEquippedItems, method getMoveSpeedMultiplier, method getArmorReduction, method recalculateInventoryWeight, method getEquipmentSummary
Detected side effects	writes diagnostics

A98. src/game/managers/ItemManager.ts

Role	Manager boundary coordinating a focused domain against Game/state.
Size	319 lines; 9955 bytes; SHA-256 prefix d267660b1ca2
Direct imports	../systems/floorItems, ../systems/stockpileZones, ../HaulManager, ../rendering/itemRenderer, ../types/items, ../types/stockpiles, ../core/utlis, ../utlis/tileAlignment
Exports	ItemManagerConfig, ItemManager
Declarations	class ItemManager, type ItemManagerConfig, method constructor, method dropItems, method pickupItems, method clearItemsAtPosition, method createStockpileZone, method removeStockpileZone, method updateStockpileItems, method requestConstructionMaterials, method isConstructionReady, method completeConstruction, method assignHaulingJob, method completeHaulingJob, method failHaulingJob, method render, method renderDebugInfo, method findBestStorageLocation, method getItemNeedingHauling, method update, method createHaulingJobForItem, method createHaulingJobsForNewZone, method cancelHaulingJobsForItem, method generateAutoHaulingJobs, method setAutoHauling
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A99. src/game/managers/MedicalManager.ts

Role	Manager boundary coordinating a focused domain against Game/state.
Size	180 lines; 6409 bytes; SHA-256 prefix 57f07807e8e8
Direct imports	../types, ../Game, ../health/medicalWorkGiver, ../health/healthSystem
Exports	MedicalManager
Declarations	class MedicalManager, method constructor, method colonistNeedsMedicalBed, method setColonistMedicalPriority, method assignMedicalTreatment, method assignComprehensiveMedicalCare, method findBestDoctor, method getColonistMedicalSkill, method treatColonist
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A100. src/game/managers/MusicManager.ts

Role	Manager boundary coordinating a focused domain against Game/state.
Size	109 lines; 3210 bytes; SHA-256 prefix ae4a091cc732
Direct imports	../AudioManager, ../audio/helpers/musicAudio

Exports	MusicManager
Declarations	class MusicManager, method stopAllMusic, method isRaidMusicActive, method isDayMusicActive
Detected side effects	writes diagnostics

A101. src/game/managers/NavigationManager.ts

Role	Manager boundary coordinating a focused domain against Game/state.
Size	468 lines; 15509 bytes; SHA-256 prefix 6a5c03deaa13
Direct imports	../Game, ../types, ../constants, ../systems/navGrid, ../core/pathfinding
Exports	NavigationManager
Declarations	class NavigationManager, method constructor, method rebuildNavGrid, method rebuildNavGridPartial, method syncTerrainToGrid, method computePathAsync, method computePath, method computePathWithDangerAvoidanceAsync, method computePathWithDangerAvoidance, method computeColonistPath, method cellIndexAt, method isBlocked, method findNearestBuilding, method findNearestTree, method findNearestRock
Detected side effects	writes diagnostics

A102. src/game/managers/RenderManager.ts

Role	Manager boundary coordinating a focused domain against Game/state.
Size	1525 lines; 53327 bytes; SHA-256 prefix e214c885884f
Direct imports	../Game, ../types, ../constants, ../render/index, ../render/debug/terrainDebugRender, ../core/particles/particleRender, ../ui/panels/colonistProfile, ../react, ../ui/contextMenu, ../ui/buildMenu, ../ui/placement, ../ui/mobilePlacement, ../ui/panels/buildingInventoryPanel, ../ui/hud/controlPanel, ../placement/placementSystem, ../buildings, ../zones, ../ui/uiUtils, ../ui/panels/performanceHUD, ../core/RenderCache
Exports	RenderManager
Declarations	class RenderManager, method constructor, method toggleWorldCache, method invalidateWorldCache, method toggleColonistCache, method toggleParticleCache, method getPerformanceStats, method render, method renderWorld, method renderMiningZones, method renderTreeZones, method renderEntities, method renderZoneDragPreview, method renderDebug, method renderNavDebug, method renderColonistDebug, method renderMiningDebug, method renderEnemyDebug, method renderUI, method drawColonistHoverTooltip, method calculateColonistSpeed
Detected side effects	writes diagnostics

A103. src/game/managers/ResearchManager.ts

Role	Manager boundary coordinating a focused domain against Game/state.
Size	285 lines; 8398 bytes; SHA-256 prefix 5912bc534c7c
Direct imports	../Game, ../research/researchDatabase, ../types, ../core/particles
Exports	ResearchProgress, ResearchManager
Declarations	class ResearchManager, type ResearchProgress, method constructor, method isCompleted, method isInProgress, method isAvailable, method getCurrentResearch, method getCurrentProgress, method startResearch, method cancelResearch, method getPartialProgress, method addProgress, method updateResearcherCount, method completeResearch, method isBuildingUnlocked, method isItemUnlocked, method isMechanicUnlocked, method getCompletedResearch, method getResearchTreeWithStatus, method serialize, method deserialize
Detected side effects	writes diagnostics

A104. src/game/managers/ReservationManager.ts

Role	Manager boundary coordinating a focused domain against Game/state.
Size	276 lines; 8054 bytes; SHA-256 prefix eabb7eeac858
Direct imports	../types, ../constants
Exports	ReservationManager
Declarations	class ReservationManager, method getAssignedTargets, method getAssignedTiles, method getInsideCount, method getBuildingCapacity, method getReservedSleepCount, method buildingHasSpace, method reserveSleepSpot, method releaseSleepReservation, method leaveBuilding, method getMaxCrew, method reserveBuildSlot, method releaseBuildReservation, method reserveTarget, method releaseTarget, method isTargetAssigned, method reserveMountainTile, method releaseMountainTile, method isMountainTileAssigned, method clearBuildingReservations, method clearAll
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A105. src/game/managers/ResourceSpawnManager.ts

Role	Manager boundary coordinating a focused domain against Game/state.
Size	150 lines; 4963 bytes; SHA-256 prefix 1eb10b0a4fde
Direct imports	../core/utills, ../constants, ../utills/tileAlignment, ../terrain, ../types
Exports	ResourceNode, ResourceSpawnManager
Declarations	class ResourceSpawnManager, type ResourceNode, method resetTimer
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A106. src/game/managers/TaskManager.ts

Role	Manager boundary coordinating a focused domain against Game/state.
Size	312 lines; 10334 bytes; SHA-256 prefix da72b3f8a080
Direct imports	../types, ../core/utills, ../systems/doorSystem, ../systems/workPriority, ../Game, ../ReservationManager
Exports	TaskManager

Declarations	class TaskManager, method clearPath, method pickTask
Detected side effects	writes diagnostics

A107. src/game/managers/UIManager.ts

Role	Manager boundary coordinating a focused domain against Game/state.
Size	357 lines; 9673 bytes; SHA-256 prefix ebf3402df0af
Direct imports	../types, ../buildings, ../ui/hud/modernHotbar, ../ui/panels/workPriorityPanel, ../ui/panels/researchPanel, /AudioManager
Exports	UIManager
Declarations	class UIManager, method selectBuilding, method toggleBuildMenu, method openBuildMenu, method closeBuildMenu, method selectColonist, method selectTargetBuilding, method deselectColonist, method toggleFollow, method cancelPendingPlacement, method startLongPress, method cancelLongPress, method isLongPressing, method setHotbarTab, method setSelectedBuildCategory, method reset
Detected side effects	schedules timers; reads/writes browser globals

A108. src/game/managers/WorkGiverManager.ts

Role	Manager boundary coordinating a focused domain against Game/state.
Size	45 lines; 1439 bytes; SHA-256 prefix 49adfb1cdc39
Direct imports	../workGivers/types, ../workGivers
Exports	WorkGiverManager
Declarations	class WorkGiverManager, method constructor, method setGivers, method getGivers, method getCandidates
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A109. src/game/placement/placementSystem.ts

Role	Supporting project script or configuration surface.
Size	450 lines; 17744 bytes; SHA-256 prefix 9a7ac912ad22
Direct imports	../buildings, ../constants, ../core/utlis, ../terrain, ../core/pathfinding, ../types, ../Game, ../audio/buildingAudioMap
Exports	canPlace, tryPlaceNow, placeAtMouse, nudgePending, rotatePending, confirmPending, cancelPending, paintPathAtMouse, paintWallAtMouse, eraseInRect, cancelOrErase, evictColonistsFrom
Declarations	fn canPlace, fn tryPlaceNow, fn placeAtMouse, fn nudgePending, fn rotatePending, fn confirmPending, fn cancelPending, fn paintPathAtMouse, fn paintWallAtMouse, fn eraseInRect, fn cancelOrErase, fn evictColonistsFrom
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A110. src/game/render/debug/heatmapDebugger.ts

Role	Canvas rendering pipeline, sprites or debug visualization.
Size	1 lines; 0 bytes; SHA-256 prefix e3b0c44298fc
Direct imports	None
Exports	None detected
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A111. src/game/render/debug/terrainDebugRender.ts

Role	Canvas rendering pipeline, sprites or debug visualization.
Size	147 lines; 5280 bytes; SHA-256 prefix 42cc0c3b974a
Direct imports	../constants, ../terrain, ../Game
Exports	drawTerrainDebug, drawDetailedTerrainDebug
Declarations	fn drawTerrainDebug, fn drawDetailedTerrainDebug
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A112. src/game/render/index.ts

Role	Canvas rendering pipeline, sprites or debug visualization.
Size	719 lines; 25759 bytes; SHA-256 prefix 70714e7ee08e
Direct imports	../constants, ../buildings, ../types, ../terrain, ../assets/images, ../colonist_systems/colonistGenerator, ../core/particles, ../systems/doorSystem, ../core/RenderOptimizations, ../core/DirtyRectTracker, ../core/RenderCache
Exports	clear, applyWorldTransform, drawGround, drawFloors, drawMountains, adjustColor, drawMountainTile, drawCircle, drawPoly, drawBuilding, drawBullets, drawBuildingProgressBars, drawHUD, type HUDData, drawPersonIcon, drawShieldIcon, drawColonistAvatar
Declarations	fn clear, fn applyWorldTransform, fn drawGround, fn drawFloors, fn drawMountains, fn adjustColor, fn drawMountainTile, fn drawCircle, fn drawPoly, fn drawBuilding, fn drawBullets, fn drawBuildingProgressBars
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A113. src/game/render/sprites/colonistRenderer.ts

Role	Canvas rendering pipeline, sprites or debug visualization.
Size	352 lines; 13232 bytes; SHA-256 prefix 9039b0356f14
Direct imports	../constants, ../assets/images, ../colonist_systems/colonistGenerator, ../core/RenderCache, ./weaponRenderer
Exports	drawColonistAvatar
Declarations	fn drawResourceGatheringProgress, fn drawTreatmentProgress, fn drawColonistAvatar
Detected side effects	reads/writes DOM; writes diagnostics

A114. src/game/render/sprites/weaponRenderer.ts

Role	Canvas rendering pipeline, sprites or debug visualization.
Size	188 lines; 6258 bytes; SHA-256 prefix ff15af816c76
Direct imports	../assets/images, ../types
Exports	drawWeapon, isMeleeWeapon, getWeaponDimensions
Declarations	fn getMeleeAnimationTransform, fn drawWeapon, fn isMeleeWeapon, fn getWeaponDimensions
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A115. src/game/rendering/itemRenderer.ts

Role	Specialized item rendering.
Size	306 lines; 10440 bytes; SHA-256 prefix 26e59053c2ae
Direct imports	../types/items, ../types/stockpiles, ../systems/floorItems
Exports	ItemRenderer
Declarations	class ItemRenderer, method constructor, method setDebugMode, method drawItemStack, method drawItemTypelcon, method getItemColor, method darkenColor, method renderHaulingPath, method highlightItem
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A116. src/game/research/researchDatabase.ts

Role	Supporting project script or configuration surface.
Size	1106 lines; 31234 bytes; SHA-256 prefix 44597bb9ea71
Direct imports	None
Exports	ResearchCategory, ResearchNode, RESEARCH_TREE, getResearchByCategory, getResearch, isResearchAvailable, getAvailableResearch, CATEGORY_INFO
Declarations	fn getResearchByCategory, fn getResearch, fn isResearchAvailable, fn getAvailableResearch, type ResearchCategory, type ResearchNode
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A117. src/game/systems/BuildingUtilities.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	1 lines; 0 bytes; SHA-256 prefix e3b0c44298fc
Direct imports	None
Exports	None detected
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A118. src/game/systems/CameraSystem.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	176 lines; 5236 bytes; SHA-256 prefix bd9ddcc260bb
Direct imports	None
Exports	Camera, CameraSystem
Declarations	class CameraSystem, type Camera, method setCanvasDimensions, method pan, method panWithSpeed, method setPosition, method centerOn, method zoom, method setZoom, method screenToWorld, method worldToScreen, method isVisible, method getCamera, method getX, method getY, method getZoom, method getCameraRef, method reset, method clampToWorld
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A119. src/game/systems/LightingSystem.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	1 lines; 0 bytes; SHA-256 prefix e3b0c44298fc
Direct imports	None
Exports	None detected
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A120. src/game/systems/ResourceSystem.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	209 lines; 5513 bytes; SHA-256 prefix 129df23e0505
Direct imports	None
Exports	Resources, ResourceType, ResourceSystem
Declarations	class ResourceSystem, type Resources, type ResourceType, method getStorageCapacity, method getTotalStored, method isStorageFull, method addResource, method subtractResource, method hasCost, method payCost, method getResource, method getAllResources, method getResourcesRef, method setResource, method reset, method formatCost
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A121. src/game/systems/TimeSystem.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	112 lines; 2618 bytes; SHA-256 prefix 5383aa87f589
Direct imports	None
Exports	TimeSystem
Declarations	class TimeSystem, method update, method isNight, method didNightJustStart, method toggleFastForward, method togglePause, method setPaused, method getEffectiveDt, method getDay, method getTimeOfDay, method getHour, method getDayLength, method getFastForward, method isPaused, method setDay, method setTimeOfDay, method setFastForward, method reset, method getCurrentHour
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A122. src/game/systems/buildingInventory.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	277 lines; 7054 bytes; SHA-256 prefix 95deeea7a0ad
Direct imports	../types
Exports	initializeBuildingInventory, getDefaultInventoryCapacity, shouldHaveInventory, addItemToInventory, removeItemFromInventory, getInventoryItemCount, hasInventorySpace, getTotalInventoryCount, getInventoryUsagePercent, clearInventory, transferItems, getResourceDisplayName, getResourceIcon
Declarations	fn initializeBuildingInventory, fn getDefaultInventoryCapacity, fn shouldHaveInventory, fn addItemToInventory, fn removeItemFromInventory, fn getInventoryItemCount, fn hasInventorySpace, fn getTotalInventoryCount, fn getInventoryUsagePercent, fn clearInventory, fn getMaxStackSize, fn transferItems, fn getResourceDisplayName, fn getResourceIcon
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A123. src/game/systems/colonistIntent.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	205 lines; 6104 bytes; SHA-256 prefix 2dd8cb40f0bc
Direct imports	../types
Exports	ColonistIntent, IntentContext, shouldEnterDecisionPhase, canInterruptColonist, forceInterruptIntent, setColonistIntent, getColonistIntent, updateColonistIntent, hasIntent, isWorking, isAvailableForTask, createWorkIntent, createRestIntent
Declarations	fn shouldEnterDecisionPhase, fn canInterruptColonist, fn forceInterruptIntent, fn setColonistIntent, fn getColonistIntent, fn updateColonistIntent, fn hasIntent, fn isWorking, fn isAvailableForTask, fn createWorkIntent, fn createRestIntent, type ColonistIntent, type IntentContext
Detected side effects	writes diagnostics

A124. src/game/systems/deferredRebuildSystem.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	93 lines; 2804 bytes; SHA-256 prefix e666aaa75892
Direct imports	../Game, ../navGrid
Exports	DeferredRebuildSystem
Declarations	class DeferredRebuildSystem, method constructor, method requestFullRebuild, method requestPartialRebuild, method processQueue, method hasQueuedRebuilds
Detected side effects	writes diagnostics

A125. src/game/systems/doorSystem.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	500 lines; 16840 bytes; SHA-256 prefix 94eb5dafb65f
Direct imports	../types
Exports	initializeDoor, isDoorPassable, isDoorBlocking, isNearDoor, requestDoorOpen, findBlockingDoor, releaseDoorQueue, updateDoor, shouldWaitAtDoor, getDoorOpenAmount, handleColonistWaitingForDoor, checkForBlockingDoorAhead, handleColonistPassingThroughDoor, debugDoorStates, attackDoor
Declarations	fn initializeDoor, fn isDoorPassable, fn isDoorBlocking, fn isNearDoor, fn requestDoorOpen, fn findBlockingDoor, fn releaseDoorQueue, fn updateDoor, fn shouldWaitAtDoor, fn getDoorOpenAmount, fn handleColonistWaitingForDoor, fn checkForBlockingDoorAhead, fn handleColonistPassingThroughDoor, fn debugDoorStates, fn attackDoor
Detected side effects	writes diagnostics

A126. src/game/systems/floorItems.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	333 lines; 8917 bytes; SHA-256 prefix 924849813b44
Direct imports	../core/utils, ../types/items
Exports	ITEM_DEFINITIONS, FloorItemManager
Declarations	class FloorItemManager, method createItem, method findStackableItemAtPosition, method stackItems, method removeItem, method takeFromItem, method getItemInArea, method getItemNearPosition, method getAllItems, method getVisualStacks, method getItemColor, method getItemWeight, method getStackLimit
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A127. src/game/systems/navGrid.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	141 lines; 5863 bytes; SHA-256 prefix c761c34d424d
Direct imports	../core/pathfinding, ../constants, ../types, ../Game, ../doorSystem
Exports	rebuildNavGridPartial, rebuildNavGrid, computePath, computePathWithDangerAvoidance, cellIndexAt, isBlocked
Declarations	fn rebuildNavGridPartial, fn rebuildNavGrid, fn computePath, fn computePathWithDangerAvoidance, fn cellIndexAt, fn isBlocked
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A128. src/game/systems/stockpileZones.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	204 lines; 6665 bytes; SHA-256 prefix 67f9d263d596
Direct imports	../core/utils, ../types/items, ../types/stockpiles, ../constants
Exports	StockpileManager
Declarations	class StockpileManager, method createZone, method removeZone, method getZone, method getAllZones, method getZoneAtPosition, method isPositionInZone, method findBestZoneForItem, method updateZoneSettings, method updateZoneAllowedItems, method getZonesByPriority
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A129. src/game/systems/treeGrowingZones.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	291 lines; 8977 bytes; SHA-256 prefix c57a84ceb9bc
Direct imports	../core/utils, ../constants
Exports	TreePlantingSpot, TreeGrowingZone, TreeGrowingManager
Declarations	class TreeGrowingManager, type TreePlantingSpot, type TreeGrowingZone, method createZone, method removeZone, method getZone, method getAllZones, method getZoneAtPosition, method isPositionInZone, method findEmptySpots, method findHarvestableSpots, method plantTree, method harvestTree, method updateGrowth, method findClosestEmptySpot, method findClosestHarvestableSpot, method updateZoneSettings, method plantTreeAt, method harvestTreeAt
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A130. src/game/systems/workPriority.ts

Role	Stateful or algorithmic domain service used by Game/managers/FSMs.
Size	403 lines; 11868 bytes; SHA-256 prefix 202de29cadcb
Direct imports	../types
Exports	WorkType, WorkPriority, DEFAULT_WORK_PRIORITIES, WorkTypeInfo, WORK_TYPE_INFO, WORK_TYPE_ORDER, canDoWorkType, getWorkPriority, initializeWorkPriorities, getWorkTypeForTask, WorkCandidate, findBestWork, setWorkPriority, cycleWorkPriority
Declarations	fn canDoWorkType, fn getWorkPriority, fn initializeWorkPriorities, fn getWorkTypeForTask, fn findBestWork, fn setWorkPriority, fn cycleWorkPriority, type WorkType, type WorkPriority, type WorkTypeInfo, type WorkCandidate
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A131. src/game/terrain.ts

Role	Supporting project script or configuration surface.
Size	748 lines; 21971 bytes; SHA-256 prefix 8ac4ede352ef
Direct imports	../constants
Exports	TerrainType, OreType, FloorType, TERRAIN_COSTS, OreProperties, ORE_PROPERTIES, FLOOR_COSTS, TerrainVisuals, TERRAIN_VISUALS, FLOOR_VISUALS, TerrainGrid, makeTerrainGrid, getTerrainType, getTerrainTypeFromId, getFloorType, getFloorTypeFromId, getOreTypeFromId, calculateMovementCost, isTerrainPassable, setTerrainRect, setFloorRect, removeFloorRect, generateTerrainFromBiome, generateMountains, updateOreVisibility, mineMountainTile, updateOreVisibilityAround, isMountainTile, getVisibleOreAt
Declarations	class SimpleNoise, fn makeTerrainGrid, fn getTerrainType, fn getTerrainTypeFromId, fn getFloorType, fn getFloorTypeFromId, fn getOreType, fn getOreTypeFromId, fn calculateMovementCost, fn isTerrainPassable, fn setTerrainRect, fn setFloorRect, fn removeFloorRect, fn generateTerrainFromBiome, fn generateMountains, fn generateOreDeposit, fn updateOreVisibility, fn mineMountainTile, fn updateOreVisibilityAround, fn isMountainTile, fn getVisibleOreAt, type TerrainType, type OreType, type FloorType, type OreProperties, type TerrainVisuals, type TerrainGrid, method constructor, method hash, method smoothstep, method noise, method octaveNoise
Detected side effects	writes diagnostics

A132. src/game/types/items.ts

Role	Compile-time contracts; erased from runtime output.
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Size	30 lines; 848 bytes; SHA-256 prefix 2be32a1c942b
Direct imports	../core/utils
Exports	ItemType, FloorItem, ItemStack, ItemDefinition
Declarations	type ItemType, type FloorItem, type ItemStack, type ItemDefinition
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A133. src/game/types/stockpiles.ts

Role	Compile-time contracts; erased from runtime output.
Size	20 lines; 354 bytes; SHA-256 prefix b25995377ada
Direct imports	./Items
Exports	StockpileZone, StockpileSettings
Declarations	type StockpileZone, type StockpileSettings
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A134. src/game/types.ts

Role	Compile-time contracts; erased from runtime output.
Size	470 lines; 20068 bytes; SHA-256 prefix f7a0ddd5dc19
Direct imports	../core/utils, ./types, ./colonist_systems/colonistGenerator
Exports	Camera, Resources, Circle, BuildingKind, BuildingDef, Building, BuildingInventoryItem, BuildingInventory, ColonistCommandIntent, ColonistState, InventoryItem, Equipment, ColonistInventory, BodyPartType, BodyPart, InjuryType, Injury, ColonistHealth, OperationType, SurgeryFailureType, Operation, ImplantType, HealthImplant, ProstheticType, HealthProsthetic, SkillName, Skill, SkillSet, Colonist, Enemy, Bullet, Particle, Message, MiningZone, Colors, World
Declarations	type Camera, type Resources, type Circle, type BuildingKind, type BuildingDef, type Building, type BuildingInventoryItem, type BuildingInventory, type ColonistCommandIntent, type ColonistState, type InventoryItem, type Equipment, type ColonistInventory, type BodyPartType, type BodyPart, type InjuryType, type Injury, type ColonistHealth, type OperationType, type SurgeryFailureType, type Operation, type ImplantType, type HealthImplant, type ProstheticType, type HealthProsthetic, type SkillName, type Skill, type SkillSet, type Colonist, type Enemy, type Bullet, type Particle, type Message, type MiningZone, type Colors, type World
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A135. src/game/ui/GameOverScreen.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	224 lines; 8241 bytes; SHA-256 prefix 8bf49438edcc
Direct imports	../Game, ../audio/helpers/musicAudio, ../react
Exports	GameOverScreen
Declarations	class GameOverScreen, type Message, type CreditEntry, method constructor, method start, method update, method render, method isActive, method stop
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A136. src/game/ui/TutorialSystem.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	1072 lines; 32424 bytes; SHA-256 prefix 908ba96a3395
Direct imports	../Game, ../audio/helpers/uiAudio, ../constants, ./contextMenu
Exports	TutorialSystem
Declarations	class TutorialSystem, type TutorialStep, method constructor, method hasCompletedBefore, method markCompleted, method start, method skip, method shouldAutoStart, method update, method updateIntro, method advanceStep, method complete, method render, method renderIntro, method renderTutorialStep, method highlightHotbarTab, method highlightElement, method drawArrow, method drawMouseWheelIcon, method measureWrappedText, method wrapTextWithTracking, method wrapText, method cleanupHighlight, method isActive, method wasSkipped, method handleClick, method shouldBlockClick, method showMobileSkipButton, method hideMobileSkipButton
Detected side effects	reads/writes DOM; persists browser state

A137. src/game/ui/bootstrap.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	156 lines; 4092 bytes; SHA-256 prefix 308ccef32ff
Direct imports	./dom/mobileControls, ../react, ../Game
Exports	UIComponents, initializeUI, linkGameToUI
Declarations	fn initializeUI, fn createWrapper, fn createCanvas, fn createLegend, fn setupKeyboardBlurring, fn linkGameToUI, type ErrorOverlayHandle, type ToastHandle, type HelpPanelHandle, type ChangelogModalHandle, type UIComponents
Detected side effects	registers event listeners; reads/writes DOM

A138. src/game/ui/buildMenu.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	151 lines; 6032 bytes; SHA-256 prefix e3327cdbe98c
Direct imports	../buildings

Exports	drawBuildMenu, handleBuildMenuClick
Declarations	fn drawBuildMenu, fn handleBuildMenuClick, fn wrapText
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A139. src/game/ui/contextMenu.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	104 lines; 3322 bytes; SHA-256 prefix 8732dd770ce2
Direct imports	./contextMenus/types, ../types, ../react
Exports	openContextMenu, hideContextMenu, isContextMenuOpen, drawLongPressProgress
Declarations	fn openContextMenu, fn hideContextMenu, fn isContextMenuOpen, fn drawLongPressProgress
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A140. src/game/ui/contextMenus/buildings/index.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	5 lines; 166 bytes; SHA-256 prefix 40d10c5201eb
Direct imports	./providers/bed, ./providers/inventory, ./providers/farm
Exports	buildingContextMenuManager, showBuildingContextMenu
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A141. src/game/ui/contextMenus/buildings/manager.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	55 lines; 1711 bytes; SHA-256 prefix bec2f807ba3c
Direct imports	../types, ../Game, ../types, ../contextMenu
Exports	BuildingContextMenuManager, buildingContextMenuManager, showBuildingContextMenu
Declarations	class BuildingContextMenuManager, fn showBuildingContextMenu, type BuildingContextMenuProvider, type ProviderMap, method register, method collectItems
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A142. src/game/ui/contextMenus/buildings/providers/bed.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	35 lines; 1095 bytes; SHA-256 prefix dbf48a97a6ce
Direct imports	../types, ../manager
Exports	None detected
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A143. src/game/ui/contextMenus/buildings/providers/farm.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	44 lines; 1528 bytes; SHA-256 prefix 5aa52bef3723
Direct imports	../types, ../manager
Exports	None detected
Declarations	type CropOption
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A144. src/game/ui/contextMenus/buildings/providers/inventory.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	33 lines; 1048 bytes; SHA-256 prefix 98813d5934f9
Direct imports	../types, ../manager, ../panels/buildingInventoryPanel
Exports	None detected
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A145. src/game/ui/contextMenus/colonistMenu.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	31 lines; 990 bytes; SHA-256 prefix a96d6e477893
Direct imports	../types, ../Game, ../types, ../contextMenu

Exports	showColonistContextMenu, buildColonistContextMenuDescriptor
Declarations	fn showColonistContextMenu, fn buildColonistContextMenuDescriptor
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A146. src/game/ui/contextMenus/types.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	30 lines; 766 bytes; SHA-256 prefix 9bcc22e19578
Direct imports	../Game
Exports	ContextMenuActionContext, ContextMenuAction, ContextMenuitem, ContextMenuDescriptor
Declarations	type ContextMenuActionContext, type ContextMenuAction, type ContextMenuitem, type ContextMenuDescriptor
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A147. src/game/ui/debugConsole.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	1523 lines; 60390 bytes; SHA-256 prefix f80543c8cf77
Direct imports	../Game, ../types, ../audio/helpers/musicAudio, ../react/stores/debugConsoleStore, ../data/itemDatabase, ../health/healthSystem, ../colonist_systems/skills, ../systems/buildingInventory, ../research/researchDatabase
Exports	DebugConsoleSystem, initDebugConsole
Declarations	class DebugConsoleSystem, fn initDebugConsole, type CommandHandler, method constructor, method register, method getCommandNames, method getCommandHelp, method execute
Detected side effects	starts intervals; writes diagnostics

A148. src/game/ui/dom/mobileControls.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	10 lines; 268 bytes; SHA-256 prefix 129ba83a63ea
Direct imports	None
Exports	MobileControlsHandle
Declarations	type MobileControlsHandle
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A149. src/game/ui/hud/controlPanel.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	216 lines; 5315 bytes; SHA-256 prefix 3dc070ff7366
Direct imports	../Game, ../audio/helpers/uiAudio
Exports	ControlPanelRect, drawControlPanel, handleControlPanelClick
Declarations	fn drawControlPanel, fn handleControlPanelClick, type ControlPanelRect
Detected side effects	reads/writes browser globals

A150. src/game/ui/hud/hotbar.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	95 lines; 2433 bytes; SHA-256 prefix 398d9292faf5
Direct imports	None
Exports	HotbarItem, HotbarRect, drawHotbar
Declarations	fn drawHotbar, fn drawHotbarSlot, type HotbarItem, type HotbarRect
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A151. src/game/ui/hud/index.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	71 lines; 1605 bytes; SHA-256 prefix 122d0c778806
Direct imports	../types, ../react, ../hotbar, ../messages, ../topBar
Exports	HUDData, drawHUD
Declarations	fn drawHUD, type HUDData
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A152. src/game/ui/hud/messages.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	55 lines; 1362 bytes; SHA-256 prefix f2b97598d4d0

Direct imports	../types
Exports	drawMessages
Declarations	fn drawMessages, fn drawMessage
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A153. src/game/ui/hud/modernBuildMenu.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	438 lines; 13163 bytes; SHA-256 prefix d2c8a8bc79b0
Direct imports	../buildings, ./modernHotbar
Exports	BuildCategory, CategoryRect, BuildingItemRect, BuildMenuRects, drawModernBuildMenu, handleBuildMenuClick
Declarations	fn clamp, fn getScaled, fn getMenuDimensions, fn drawModernBuildMenu, fn handleBuildMenuClick, type BuildCategory, type CategoryRect, type BuildingItemRect, type BuildMenuRects
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A154. src/game/ui/hud/modernHotbar.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	180 lines; 5432 bytes; SHA-256 prefix 499b3283f5fd
Direct imports	None
Exports	HotbarTab, HotbarTabRect, getModernHotbarHeight, drawModernHotbar, handleHotbarClick
Declarations	fn getModernHotbarHeight, fn drawModernHotbar, fn handleHotbarClick, type HotbarTab, type HotbarTabRect
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A155. src/game/ui/hud/modernWorkPanel.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	57 lines; 1628 bytes; SHA-256 prefix 7dd7a5213105
Direct imports	./modernHotbar
Exports	getWorkPanelLayout, drawWorkPanelContainer
Declarations	fn getWorkPanelLayout, fn drawWorkPanelContainer
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A156. src/game/ui/hud/topBar.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	232 lines; 6727 bytes; SHA-256 prefix 7aa87ac07b20
Direct imports	None
Exports	TopBarData, drawTopBar
Declarations	fn drawTopBar, fn getPillHeight, fn measurePillWidth, fn measureBlockWidth, fn drawPill, fn drawRoundedRect, fn formatValue, type TopBarData, type PillRenderOptions
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A157. src/game/ui/mobilePlacement.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	308 lines; 8522 bytes; SHA-256 prefix d22623d986fc
Direct imports	../buildings, ../placement/placementSystem, ../constants, ../Game
Exports	drawMobilePlacementUI, handleMobilePlacementClick, isClickOnGhost
Declarations	fn drawMobilePlacementUI, fn createPlacementButtons, fn drawPlacementButtons, fn drawInstructionText, fn handleMobilePlacementClick, fn handleClick, fn cleanupPlacementUI, fn isClickOnGhost, type PlacementButton
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A158. src/game/ui/panels/buildingInventoryPanel.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	270 lines; 7633 bytes; SHA-256 prefix 6259c4dd7cd1
Direct imports	../types, ../systems/buildingInventory
Exports	openBuildingInventoryPanel, closeBuildingInventoryPanel, isBuildingInventoryPanelOpen, getSelectedBuilding, drawBuildingInventoryPanel, handleBuildingInventoryPanelClick
Declarations	fn openBuildingInventoryPanel, fn closeBuildingInventoryPanel, fn isBuildingInventoryPanelOpen, fn getSelectedBuilding, fn drawBuildingInventoryPanel, fn handleBuildingInventoryPanelClick
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A159. src/game/ui/panels/colonistProfile.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	608 lines; 25177 bytes; SHA-256 prefix c79d0a451b7f
Direct imports	../././react, ../././render
Exports	drawColonistProfile, drawSkillsTab, drawSocialTab, drawLogTab, measureSkillsTabHeight, measureSocialTabHeight, measureLogTabHeight
Declarations	fn drawColonistProfile, fn syncColonistProfilePanel, fn drawBioTab, fn drawSkillsTab, fn drawSocialTab, fn drawLogTab, fn formatLogTime, fn getLogColor, fn measureSkillsTabHeight, fn measureSocialTabHeight, fn measureLogTabHeight, type ActivityLogEntry
Detected side effects	reads/writes DOM

A160. src/game/ui/panels/healthTab.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	1060 lines; 38979 bytes; SHA-256 prefix 38f0ad07716a
Direct imports	.././types, .././health/healthSystem
Exports	drawHealthTab, measureHealthTabHeight
Declarations	fn drawHealthTab, fn measureHealthTabHeight, fn measureOverviewHeight, fn measureOperationsHeight, fn measureInjuriesHeight, fn drawOverviewTab, fn drawOperationsTab, fn drawInjuriesTab, fn calculateSystemCapacity, fn getCapacityColor, fn formatBodyPartLabel, fn getAvailableOperations, fn getOperationIcon, type BodySystem, type HealthTabContext
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A161. src/game/ui/panels/performanceHUD.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	373 lines; 11621 bytes; SHA-256 prefix 1637efd7fba1
Direct imports	../././core/PerformanceMetrics, ../././core/SimulationClock, ../././core/BudgetedExecution, ../././Game
Exports	PerformanceHUDConfig, PerformanceHUD, initPerformanceHUD, getPerformanceHUD, togglePerformanceHUD, drawPerformanceHUD
Declarations	class PerformanceHUD, fn initPerformanceHUD, fn getPerformanceHUD, fn togglePerformanceHUD, fn drawPerformanceHUD, type PerformanceHUDConfig, method constructor, method toggle, method setVisible, method configure, method draw, method updateOffscreenCanvas, method blitToScreen, method buildHUDContent, method createBar, method getConfig
Detected side effects	reads/writes DOM

A162. src/game/ui/panels/researchPanel.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	64 lines; 1677 bytes; SHA-256 prefix 6d49218840f1
Direct imports	../././react/stores/researchStore
Exports	toggleResearchPanel, isResearchPanelOpen, closeResearchPanel, openResearchPanel
Declarations	fn toggleResearchPanel, fn isResearchPanelOpen, fn closeResearchPanel, fn openResearchPanel
Detected side effects	reads/writes DOM

A163. src/game/ui/panels/workPriorityPanel.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	49 lines; 1292 bytes; SHA-256 prefix db15d956aef
Direct imports	../././react/stores/workPriorityStore, ../././react/stores/mobileControlsStore
Exports	toggleWorkPriorityPanel, isWorkPriorityPanelOpen, closeWorkPriorityPanel
Declarations	fn toggleWorkPriorityPanel, fn isWorkPriorityPanelOpen, fn closeWorkPriorityPanel
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A164. src/game/ui/placement.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	55 lines; 3411 bytes; SHA-256 prefix 577dd71749e3
Direct imports	../buildings, ../placement/placementSystem, ../constants
Exports	drawPlacementUI
Declarations	fn drawPlacementUI
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A165. src/game/ui/renderUtils.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	90 lines; 2262 bytes; SHA-256 prefix f0b1cfc44861
Direct imports	../constants
Exports	drawPersonIcon, drawShieldIcon
Declarations	fn drawPersonIcon, fn drawShieldIcon

Detected side effects	none detected by static scan; behavior is exposed through imports/exports
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A166. src/game/ui/uiUtils.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	92 lines; 2071 bytes; SHA-256 prefix a61641b6d421
Direct imports	None
Exports	drawTooltip, isPointInCircle, isPointInRect
Declarations	fn drawTooltip, fn isPointInCircle, fn isPointInRect
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A167. src/game/utils/tileAlignment.ts

Role	Supporting project script or configuration surface.
Size	112 lines; 3420 bytes; SHA-256 prefix 2707be1a10a1
Direct imports	../constants, ../../core/utils
Exports	snapToTileCenter, getTileCenterPos, worldToGrid, isAlignedToTileCenter, findNearbyAlignedPosition
Declarations	fn snapToTileCenter, fn getTileCenterPos, fn worldToGrid, fn isAlignedToTileCenter, fn findNearbyAlignedPosition
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A168. src/game/workGivers/construction.ts

Role	Job eligibility, scoring and assignment logic consumed by work orchestration.
Size	36 lines; 1085 bytes; SHA-256 prefix 51d09fa4b068
Direct imports	./types, ../types
Exports	ConstructionWorkGiver
Declarations	method getCandidates
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A169. src/game/workGivers/cooking.ts

Role	Job eligibility, scoring and assignment logic consumed by work orchestration.
Size	55 lines; 1934 bytes; SHA-256 prefix 102d1a33d65e
Direct imports	./types
Exports	CookingWorkGiver
Declarations	method getCandidates
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A170. src/game/workGivers/cooling.ts

Role	Job eligibility, scoring and assignment logic consumed by work orchestration.
Size	71 lines; 2579 bytes; SHA-256 prefix 8ccd24322da2
Direct imports	./types
Exports	CoolingWorkGiver
Declarations	method getCandidates
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A171. src/game/workGivers/equipment.ts

Role	Job eligibility, scoring and assignment logic consumed by work orchestration.
Size	97 lines; 3176 bytes; SHA-256 prefix 367744b918e9
Direct imports	./types, ../colonist_systems/states/equipmentState
Exports	EquipmentWorkGiver
Declarations	fn isWeapon, fn getWeaponValue, method getCandidates
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A172. src/game/workGivers/floorHauling.ts

Role	Job eligibility, scoring and assignment logic consumed by work orchestration.
Size	51 lines; 1793 bytes; SHA-256 prefix ccc44aa16422
Direct imports	./types, ../types/items
Exports	FloorHaulingWorkGiver
Declarations	method getCandidates

Detected side effects	none detected by static scan; behavior is exposed through imports/exports
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A173. src/game/workGivers/growing.ts

Role	Job eligibility, scoring and assignment logic consumed by work orchestration.
Size	30 lines; 929 bytes; SHA-256 prefix 0b6391abc175
Direct imports	./types
Exports	GrowingWorkGiver
Declarations	method getCandidates
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A174. src/game/workGivers/hauling.ts

Role	Job eligibility, scoring and assignment logic consumed by work orchestration.
Size	53 lines; 2301 bytes; SHA-256 prefix 9fad409c6740
Direct imports	./types, ../systems/buildingInventory
Exports	HaulingWorkGiver
Declarations	method getCandidates
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A175. src/game/workGivers/index.ts

Role	Job eligibility, scoring and assignment logic consumed by work orchestration.
Size	40 lines; 1681 bytes; SHA-256 prefix 82f0814d8fbb
Direct imports	./types, ./growing, ./waterCollection, ./construction, ./plantCutting, ./mining, ./cooking, ./stonecutting, ./smelting, ./cooling, ./floorHauling, ./research, ./treePlanting, ./smithing, ./equipment, ./patient, ./medical
Exports	WORK_GIVERS
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A176. src/game/workGivers/medical.ts

Role	Job eligibility, scoring and assignment logic consumed by work orchestration.
Size	64 lines; 2041 bytes; SHA-256 prefix d248fbab6572
Direct imports	./types, ../health/medicalWorkGiver
Exports	MedicalWorkGiver
Declarations	method getCandidates
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A177. src/game/workGivers/mining.ts

Role	Job eligibility, scoring and assignment logic consumed by work orchestration.
Size	134 lines; 5128 bytes; SHA-256 prefix 0cdcb683853
Direct imports	./types, ../terrain, ../constants, ../utils/tileAlignment
Exports	MiningWorkGiver
Declarations	method getCandidates
Detected side effects	writes diagnostics

A178. src/game/workGivers/patient.ts

Role	Job eligibility, scoring and assignment logic consumed by work orchestration.
Size	87 lines; 3587 bytes; SHA-256 prefix a5ff0e6bb7d6
Direct imports	./types, ../types
Exports	PatientWorkGiver
Declarations	method getCandidates
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A179. src/game/workGivers/plantCutting.ts

Role	Job eligibility, scoring and assignment logic consumed by work orchestration.
Size	28 lines; 983 bytes; SHA-256 prefix 300c7477a9ec
Direct imports	./types
Exports	PlantCuttingWorkGiver

Declarations	method getCandidates
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A180. src/game/workGivers/research.ts

Role	Job eligibility, scoring and assignment logic consumed by work orchestration.
Size	46 lines; 1342 bytes; SHA-256 prefix c0203136650f
Direct imports	./types, ../types
Exports	ResearchWorkGiver
Declarations	method getCandidates
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A181. src/game/workGivers/smelting.ts

Role	Job eligibility, scoring and assignment logic consumed by work orchestration.
Size	61 lines; 2264 bytes; SHA-256 prefix 9adaf5901c27
Direct imports	./types
Exports	SmeltingWorkGiver
Declarations	method getCandidates
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A182. src/game/workGivers/smithing.ts

Role	Job eligibility, scoring and assignment logic consumed by work orchestration.
Size	157 lines; 5570 bytes; SHA-256 prefix 9e20f3a4d81b
Direct imports	./types
Exports	SmithingWorkGiver
Declarations	fn getCraftableWeapons, fn countExistingWeapons, method getCandidates
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A183. src/game/workGivers/stonecutting.ts

Role	Job eligibility, scoring and assignment logic consumed by work orchestration.
Size	60 lines; 2200 bytes; SHA-256 prefix 03d37a13cc33
Direct imports	./types
Exports	StonecuttingWorkGiver
Declarations	method getCandidates
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A184. src/game/workGivers/treePlanting.ts

Role	Job eligibility, scoring and assignment logic consumed by work orchestration.
Size	57 lines; 1855 bytes; SHA-256 prefix 136dd459af0c
Direct imports	./types
Exports	TreePlantingWorkGiver
Declarations	method getCandidates
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A185. src/game/workGivers/types.ts

Role	Job eligibility, scoring and assignment logic consumed by work orchestration.
Size	21 lines; 470 bytes; SHA-256 prefix c3878f64b7b5
Direct imports	None
Exports	WorkCandidate, WorkGiverContext, WorkGiver
Declarations	type WorkCandidate, type WorkGiverContext, type WorkGiver
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A186. src/game/workGivers/waterCollection.ts

Role	Job eligibility, scoring and assignment logic consumed by work orchestration.
Size	41 lines; 1393 bytes; SHA-256 prefix da1206cc454c
Direct imports	./types
Exports	WaterCollectionWorkGiver

Declarations	method getCandidates
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A187. src/game/zones.ts

Role	Supporting project script or configuration surface.
Size	61 lines; 1634 bytes; SHA-256 prefix b25a08ebbeb3
Direct imports	./constants
Exports	ZoneDef, ZONE_TYPES, isZone, getZoneDef
Declarations	fn isZone, fn getZoneDef, type ZoneDef
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A188. src/main.ts

Role	Browser entry; bootstraps UI, assets, Game, React, globals and error reporting.
Size	128 lines; 4017 bytes; SHA-256 prefix 4a9e13e044d7
Direct imports	./game/Game, ./game/buildings, ./assets/images, ./game/ui/bootstrap, ./react/main
Exports	None detected
Declarations	fn initGame, fn createLoadingOverlay, fn createLoadingUpdater, fn formatAssetLabel, type LoadingOverlayElements
Detected side effects	reads/writes DOM; writes diagnostics

A189. src/react/gameOver/GameOverOverlay.tsx

Role	React UI component, external store or imperative bridge.
Size	229 lines; 7823 bytes; SHA-256 prefix 552bfc7a6b6e
Direct imports	react, .././game/render/sprites/colonistRenderer, ../stores/gameOverStore
Exports	GameOverOverlay
Declarations	fn MemorialCard, fn GameOverOverlay, type MemorialCardProps
Detected side effects	writes diagnostics

A190. src/react/index.ts

Role	React UI component, external store or imperative bridge.
Size	43 lines; 2888 bytes; SHA-256 prefix f8d870dd282e
Direct imports	None
Exports	GameOverOverlay, TopBar, MobileControls, ResourceBar, Hotbar, BuildMenu, ColonistProfilePanel, WorkPriorityPanel, ResearchPanel, DebugConsole, HelpPanel, ChangelogModal, Toast, ErrorOverlay, ContextMenu, createMobileControlsBridge, type MobileControlsHandle, createHelpPanelBridge, type HelpPanelHandle, createChangelogBridge, type ChangelogModalHandle, createToastBridge, type ToastHandle, createErrorOverlayBridge, type ErrorOverlayHandle, loadChangelogEntries, sanitizeChangelogContent, getMobileControlsState, subscribeMobileControls, setMobileControlsState, getHelpPanelState, subscribeHelpPanel, setHelpPanelVisible, toggleHelpPanel, getChangelogState, subscribeChangelog, setChangelogState, setChangelogVisible, getToastState, subscribeToast, showToast, hideToast, getErrorOverlayState, subscribeErrorOverlay, showErrorOverlay, clearErrorOverlay, getContextMenuState, subscribeContextMenu, showContextMenu, hideContextMenuState, getResourceBarState, subscribeResourceBar, setResourceBarState, getHotbarState, subscribeHotbar, setHotbarState, getColonistProfileState, subscribeColonistProfile, setColonistProfileState, getWorkPriorityState, subscribeWorkPriority, setWorkPriorityVisible, toggleWorkPriorityPanel, getResearchPanelState, subscribeResearchPanel, setResearchPanelVisible, toggleResearchPanel, getDebugConsoleState, subscribeDebugConsole, setDebugConsoleOpen, getGameOverOverlayState, getGameOverOverlayState, subscribeGameOverOverlay, setGameOverOverlayState, resetGameOverOverlayState, type GameOverOverlayState, type GuiltMessage, type CreditEntry
Declarations	type GameOverOverlayState, type GuiltMessage, type CreditEntry
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A191. src/react/main.tsx

Role	React UI component, external store or imperative bridge.
Size	70 lines; 1875 bytes; SHA-256 prefix cd2971f40d83
Direct imports	react-dom/client, ./index, ./ui/TopBar, ./ui/MobileControls, ./ui/ResourceBar, ./ui/Hotbar, ./ui/BuildMenu, ./ui/ColonistBioPanel, ./ui/WorkPriorityPanel, ./ui/ResearchPanel, ./ui/HelpPanel, ./ui/ChangelogModal, ./ui/Toast, ./ui/ErrorOverlay, ./ui/ContextMenu
Exports	None detected
Declarations	None detected
Detected side effects	schedules animation frames; reads/writes DOM

A192. src/react/stores/changelogStore.ts

Role	React UI component, external store or imperative bridge.
Size	38 lines; 901 bytes; SHA-256 prefix e43f8ea74bd5
Direct imports	./changelogTypes
Exports	ChangelogState, getChangelogState, subscribeChangelog, setChangelogState, setChangelogVisible
Declarations	fn getChangelogState, fn subscribeChangelog, fn setChangelogState, fn setChangelogVisible, type ChangelogState
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A193. src/react/stores/changelogTypes.ts

Role	React UI component, external store or imperative bridge.
Size	17 lines; 279 bytes; SHA-256 prefix 455373aa5922
Direct imports	None
Exports	CommitInfo, ChangelogEntry
Declarations	type CommitInfo, type ChangelogEntry
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A194. src/react/stores/colonistBioStore.ts

Role	React UI component, external store or imperative bridge.
Size	120 lines; 2782 bytes; SHA-256 prefix 41590d5a6013
Direct imports	None
Exports	ColonistBioFamily, ColonistBioProfile, ColonistSkill, ColonistGearItem, ColonistGearEquipment, ColonistGearState, ColonistSocialEntry, ColonistLogEntry, ColonistProfileState, getColonistProfileState, subscribeColonistProfile, setColonistProfileState
Declarations	fn getColonistProfileState, fn subscribeColonistProfile, fn setColonistProfileState, type ColonistBioFamily, type ColonistBioProfile, type ColonistSkill, type ColonistGearItem, type ColonistGearEquipment, type ColonistGearState, type ColonistSocialEntry, type ColonistLogEntry, type ColonistProfileState
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A195. src/react/stores/contextMenuStore.ts

Role	React UI component, external store or imperative bridge.
Size	72 lines; 1587 bytes; SHA-256 prefix b03b13965dbd
Direct imports	../game/ui/contextMenus/types
Exports	ContextMenuState, getContextMenuState, subscribeContextMenu, setContextMenuState, showContextMenu, hideContextMenuState
Declarations	fn getContextMenuState, fn subscribeContextMenu, fn setContextMenuState, fn showContextMenu, fn hideContextMenuState, type ContextMenuState
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A196. src/react/stores/debugConsoleStore.ts

Role	React UI component, external store or imperative bridge.
Size	46 lines; 1173 bytes; SHA-256 prefix b4278eea658b
Direct imports	None
Exports	DebugConsoleState, getDebugConsoleState, subscribeDebugConsole, setDebugConsoleState, setDebugConsoleOpen, appendDebugConsoleOutput
Declarations	fn getDebugConsoleState, fn subscribeDebugConsole, fn setDebugConsoleState, fn setDebugConsoleOpen, fn appendDebugConsoleOutput, type DebugConsoleState
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A197. src/react/stores/errorOverlayStore.ts

Role	React UI component, external store or imperative bridge.
Size	35 lines; 857 bytes; SHA-256 prefix d3a4e84a43c1
Direct imports	None
Exports	ErrorOverlayState, getErrorOverlayState, subscribeErrorOverlay, setErrorOverlayState, showErrorOverlay, clearErrorOverlay
Declarations	fn getErrorOverlayState, fn subscribeErrorOverlay, fn setErrorOverlayState, fn showErrorOverlay, fn clearErrorOverlay, type ErrorOverlayState
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A198. src/react/stores/gameOverStore.ts

Role	React UI component, external store or imperative bridge.
Size	72 lines; 1521 bytes; SHA-256 prefix 84930b22f7b3
Direct imports	None
Exports	CreditEntry, GuiltMessage, GameOverOverlayState, getGameOverOverlayState, subscribeGameOverOverlay, setGameOverOverlayState, resetGameOverOverlayState
Declarations	fn getGameOverOverlayState, fn subscribeGameOverOverlay, fn setGameOverOverlayState, fn resetGameOverOverlayState, type CreditEntry, type GuiltMessage, type GameOverOverlayState
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A199. src/react/stores/helpPanelStore.ts

Role	React UI component, external store or imperative bridge.
Size	29 lines; 636 bytes; SHA-256 prefix fc8903c61b1d
Direct imports	None
Exports	HelpPanelState, getHelpPanelState, subscribeHelpPanel, setHelpPanelVisible, toggleHelpPanel

Declarations	fn getHelpPanelState, fn subscribeHelpPanel, fn setHelpPanelVisible, fn toggleHelpPanel, type HelpPanelState
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A200. src/react/stores/hotbarStore.ts

Role	React UI component, external store or imperative bridge.
Size	121 lines; 3527 bytes; SHA-256 prefix b86fad2b4e31
Direct imports	../game/ui/hud/modernHotbar
Exports	HotbarState, getHotbarState, subscribeHotbar, setHotbarState, ensureHotbarViewportTracking
Declarations	fn readSafeArealnsetBottom, fn readViewportMetrics, fn getHotbarState, fn subscribeHotbar, fn setHotbarState, fn ensureHotbarViewportTracking, type HotbarState
Detected side effects	registers event listeners; reads/writes DOM; reads/writes browser globals

A201. src/react/stores/mobileControlsStore.ts

Role	React UI component, external store or imperative bridge.
Size	40 lines; 1009 bytes; SHA-256 prefix e63b74c46a45
Direct imports	None
Exports	MobileControlsState, getMobileControlsState, subscribeMobileControls, setMobileControlsState, setMobileControlsSuppressed
Declarations	fn getMobileControlsState, fn subscribeMobileControls, fn setMobileControlsState, fn setMobileControlsSuppressed, type MobileControlsState
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A202. src/react/stores/researchStore.ts

Role	React UI component, external store or imperative bridge.
Size	38 lines; 761 bytes; SHA-256 prefix 56ec59a9015e
Direct imports	None
Exports	subscribeResearchPanel, getResearchPanelState, setResearchPanelVisible, toggleResearchPanel
Declarations	fn notifyListeners, fn subscribeResearchPanel, fn getResearchPanelState, fn setResearchPanelVisible, fn toggleResearchPanel, type ResearchPanelState
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A203. src/react/stores/resourceBarStore.ts

Role	React UI component, external store or imperative bridge.
Size	49 lines; 949 bytes; SHA-256 prefix ba43c0f4a98b
Direct imports	None
Exports	ResourceBarState, getResourceBarState, subscribeResourceBar, setResourceBarState
Declarations	fn getResourceBarState, fn subscribeResourceBar, fn setResourceBarState, type ResourceBarState
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A204. src/react/stores/toastStore.ts

Role	React UI component, external store or imperative bridge.
Size	50 lines; 1069 bytes; SHA-256 prefix fdd3832c851c
Direct imports	None
Exports	ToastState, getToastState, subscribeToast, setToastState, showToast, hideToast
Declarations	fn getToastState, fn subscribeToast, fn setToastState, fn showToast, fn hideToast, type ToastState
Detected side effects	schedules timers; reads/writes browser globals

A205. src/react/stores/workPriorityStore.ts

Role	React UI component, external store or imperative bridge.
Size	38 lines; 760 bytes; SHA-256 prefix e60bfd7e7ae7
Direct imports	None
Exports	subscribeWorkPriority, getWorkPriorityState, setWorkPriorityVisible, toggleWorkPriorityPanel
Declarations	fn notifyListeners, fn subscribeWorkPriority, fn getWorkPriorityState, fn setWorkPriorityVisible, fn toggleWorkPriorityPanel, type WorkPriorityState
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A206. src/react/ui/BuildMenu.tsx

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	177 lines; 7784 bytes; SHA-256 prefix ab67334fcb19
Direct imports	react, ../game/buildings, ../stores/hotbarStore
Exports	BuildMenu

Declarations	fn clamp, fn getMenuDimensions, fn getHotbarHeight, fn BuildMenu
Detected side effects	reads/writes browser globals

A207. src/react/ui/ChangelogModal.tsx

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	118 lines; 4628 bytes; SHA-256 prefix 955085b80e27
Direct imports	react, ../stores/changelogStore, ./changelogData, ../stores/changelogTypes
Exports	ChangelogModal
Declarations	fn ChangelogModal
Detected side effects	registers event listeners; reads/writes DOM; writes diagnostics

A208. src/react/ui/ColonistBioPanel.tsx

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	431 lines; 15654 bytes; SHA-256 prefix 2ff27335613c
Direct imports	react, ../game/render/index, ../game/audio/helpers/uiAudio, ../stores/colonistBioStore, ./ColonistProfileTabs
Exports	ColonistProfilePanel
Declarations	fn Tooltip, fn getTraitTooltip, fn getSkillTooltip, fn ColonistProfilePanel, fn BioHeader, fn BioTab, fn BioSection, fn renderFamily, type TooltipProps
Detected side effects	reads/writes browser globals

A209. src/react/ui/ColonistProfileTabs.tsx

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	533 lines; 21123 bytes; SHA-256 prefix f6c3c2c148f1
Direct imports	react
Exports	SkillsTab, GearTab, SocialTab, LogTab, HealthTab
Declarations	fn SkillsTab, fn GearTab, fn SocialTab, fn LogTab, fn HealthTab, fn HealthOverviewTab, fn HealthOperationsTab, fn HealthInjuriesTab
Detected side effects	starts intervals

A210. src/react/ui/ContextMenu.tsx

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	129 lines; 4676 bytes; SHA-256 prefix 31a9ff6ec103
Direct imports	react, ../stores/contextMenuStore, ../game/ui/contextMenus/types
Exports	ContextMenu
Declarations	fn ContextMenu, fn clamp
Detected side effects	registers event listeners; schedules animation frames; reads/writes DOM; reads/writes browser globals

A211. src/react/ui/DebugConsole.tsx

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	213 lines; 6536 bytes; SHA-256 prefix f60589f5ff5d
Direct imports	react, ../stores/debugConsoleStore
Exports	DebugConsole
Declarations	fn DebugConsole
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A212. src/react/ui/ErrorOverlay.tsx

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	60 lines; 1525 bytes; SHA-256 prefix b89251add846
Direct imports	react, ../stores/errorOverlayStore
Exports	ErrorOverlay
Declarations	fn ErrorOverlay
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A213. src/react/ui/HelpPanel.tsx

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	69 lines; 2764 bytes; SHA-256 prefix 915b55760e19
Direct imports	react, ../stores/helpPanelStore
Exports	HelpPanel

Declarations	fn HelpPanel
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A214. src/react/ui/Hotbar.tsx

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	98 lines; 3386 bytes; SHA-256 prefix 981822ca8c5b
Direct imports	react, ../../game/ui/hud/modernHotbar, ../stores/hotbarStore
Exports	Hotbar
Declarations	fn getHotbarHeight, fn Hotbar
Detected side effects	reads/writes browser globals

A215. src/react/ui/MobileControls.tsx

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	110 lines; 3099 bytes; SHA-256 prefix 8b7273d871bf
Direct imports	react, ../stores/mobileControlsStore
Exports	MobileControls
Declarations	fn MobileControls
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A216. src/react/ui/ResearchPanel.tsx

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	421 lines; 15430 bytes; SHA-256 prefix 0d21b2166afd
Direct imports	react, ../../game/research/researchDatabase, ../stores/researchStore, ../stores/mobileControlsStore
Exports	ResearchPanel
Declarations	fn ResearchPanel, type TooltipState, type LockedPopupState
Detected side effects	starts intervals

A217. src/react/ui/ResourceBar.tsx

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	161 lines; 5805 bytes; SHA-256 prefix 8b098c0ce7a8
Direct imports	react, ../stores/resourceBarStore
Exports	ResourceBar
Declarations	fn ResourceBar, fn formatValue
Detected side effects	registers event listeners; reads/writes DOM

A218. src/react/ui/Toast.tsx

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	19 lines; 434 bytes; SHA-256 prefix 14bc691407dd
Direct imports	react, ../stores/toastStore
Exports	Toast
Declarations	fn Toast
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A219. src/react/ui/TopBar.tsx

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	9 lines; 132 bytes; SHA-256 prefix bc7326964dcb
Direct imports	None
Exports	TopBar
Declarations	fn TopBar
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A220. src/react/ui/WorkPriorityPanel.tsx

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	164 lines; 6023 bytes; SHA-256 prefix f1e80f9076b
Direct imports	react, ../stores/workPriorityStore, ../../game/systems/workPriority, ../../game/types
Exports	WorkPriorityPanel

Declarations	fn WorkPriorityPanel
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A221. src/react/ui/changelogBridge.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	23 lines; 526 bytes; SHA-256 prefix 6443279ff3e1
Direct imports	../stores/changelogStore, ./changelogData
Exports	ChangelogModalHandle, createChangelogBridge
Declarations	fn createChangelogBridge, type ChangelogModalHandle, method show, method hide, method sanitizeContent
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A222. src/react/ui/changelogData.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	224 lines; 7278 bytes; SHA-256 prefix 35606cd5158a
Direct imports	../stores/changelogTypes
Exports	sanitizeChangelogContent
Declarations	fn loadChangelogEntries, fn sanitizeChangelogContent, fn sanitizeCommitMessage, fn sanitizeAuthor, fn fetchFromGitHub, fn fetchCommitsForRelease, fn fetchFromLocal, fn parseLocalMarkdown, fn parseMarkdown
Detected side effects	writes diagnostics

A223. src/react/ui/errorOverlayBridge.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	68 lines; 1900 bytes; SHA-256 prefix a667c6e23902
Direct imports	../stores/errorOverlayStore
Exports	ErrorOverlayHandle, createErrorOverlayBridge
Declarations	fn ensureErrorHandlers, fn createErrorOverlayBridge, type ErrorOverlayHandle, method show, method clear
Detected side effects	registers event listeners; reads/writes browser globals; writes diagnostics

A224. src/react/ui/helpPanelBridge.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	22 lines; 400 bytes; SHA-256 prefix dcdc821e19d4
Direct imports	../stores/helpPanelStore
Exports	HelpPanelHandle, createHelpPanelBridge
Declarations	fn createHelpPanelBridge, type HelpPanelHandle, method show, method hide, method toggle
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A225. src/react/ui/mobileControlsBridge.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	38 lines; 1056 bytes; SHA-256 prefix edd975f9f253
Direct imports	../stores/mobileControlsStore
Exports	MobileControlsHandle, createMobileControlsBridge
Declarations	fn createMobileControlsBridge, type MobileControlsHandle, method show, method hide, method setPauseState, method setFastForwardState, method setEraseState, method showSkipTutorialButton, method hideSkipTutorialButton
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A226. src/react/ui/toastBridge.ts

Role	Imperative canvas/DOM UI, input adapters, panels or bridges.
Size	18 lines; 359 bytes; SHA-256 prefix c00cc36f5ef4
Direct imports	../stores/toastStore
Exports	ToastHandle, createToastBridge
Declarations	fn createToastBridge, type ToastHandle, method show, method hide
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A227. src/types/images.d.ts

Role	Compile-time contracts; erased from runtime output.
Size	2 lines; 11 bytes; SHA-256 prefix 8e609bb71c20
Direct imports	None

Exports	None detected
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A228. tsconfig.json

Role	Supporting project script or configuration surface.
Size	17 lines; 394 bytes; SHA-256 prefix 3c5cec86fd9e
Direct imports	None
Exports	None detected
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

A229. vite.config.ts

Role	Vite build/dev/asset/worker configuration.
Size	28 lines; 718 bytes; SHA-256 prefix c85c3fc18690
Direct imports	vite, @vitejs/plugin-react
Exports	None detected
Declarations	None detected
Detected side effects	none detected by static scan; behavior is exposed through imports/exports

Appendix B. Coverage and Limitations

- Cataloged 229 script/configuration surfaces totaling 50,991 lines.
- Included all repository .ts, .tsx, .js and .sh files outside node_modules/.git, plus package.json, tsconfig.json, index.html and the deployment workflow.
- Binary PNG/PSD/OGG assets are consumed through images.ts and audio/manifest.ts; individual binaries do not contain executable code and are not cataloged as scripts.
- medicalSystem.ts.bak is a backup artifact, not compiled by normal TypeScript imports; remove it after confirming Git history contains the needed version.
- Static regex extraction can miss computed/dynamic relationships and may classify methods conservatively. The architectural sections account for known callback, FSM and bridge flows; use browser profiling and coverage for executed-path proof.
- No server/back-end code exists in this repository snapshot; back-end recommendations are prospective.